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Neurosymbolic Large Language Models: A Survey of Symbolic Integration, Reasoning and Explainability

Maneeha Rani[1] · **Bhupesh Kumar Mishra**[1] · **Dhavalkumar Thakker**[2]

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Abstract
LLMs have demonstrated strong language-learning and human-like response-generation capabilities, and they are increasingly used to support decision-making in high-risk sectors. However, their internal decision processes remain difficult to interpret, and their responses may lack transparency. The literature has explored numerous approaches to address transparency challenges in LLMs, including Neurosymbolic AI (NeSy AI). NeSy AI approaches were primarily developed for conventional neural networks and may not transfer directly to the distinctive characteristics of LLMs. Consequently, there is a limited systematic understanding of how symbolic AI can be effectively integrated into LLMs. This paper aims to address this gap by first reviewing established NeSy AI methods and then proposing a novel taxonomy of symbolic integration in LLMs, along with a roadmap to merge symbolic techniques with LLMs. The taxonomy organises the literature across four dimensions: (1) the stage of LLM development at which symbolic information is integrated; (2) the coupling mechanism; (3) the architectural paradigm; and (4) the algorithm-level or application-level perspective. The review identifies commonly used benchmarks, recent advances and important research gaps, and uses these findings to outline directions for future research. By highlighting the latest developments and notable gaps in the literature, it offers practical insights for implementing frameworks for symbolic integration into LLMs to enhance transparency.

Keywords Neurosymbolic AI · Large language models · Symbolic integration · Knowledge graphs · Reasoning · Explainability

Abbreviations

Abbreviation	Full Form
NeSy AI	Neurosymbolic Artificial Intelligence
ARC	Automated Reasoning Critic | Abstraction and Reasoning Corpus
ARB	Advanced Reasoning Benchmark
ASP	Answer Set Programming
ATOMIC	ATlas of MachIne Commonsense
ChatKBQA	Chat for Knowledge Base Question Answering
CodeXGLUE	Code-based General Language Understanding Evaluation Benchmark
CoK	Chain-of-Knowledge
CoLAKE	Contextualized Language and Knowledge Embedding
COMET	Commonsense Transformers for Automatic Knowledge Graph Construction
CoT	Chain of Thoughts
CWQ	Complex WebQuestions
DKPLM	Decomposable Knowledge-Enhanced Pretrained Language Model
DocRED	Document-Level Relation Extraction Dataset
ERNIE	Enhanced Language Representation with Informative Entities
FOL	First Order Logic
FOLIO	First-Order Logic Inference
GAIN	Graph Aggregation-and-Inference

✉ Maneeha Rani
maneeharani@gmail.com; m.rani3-2022@hull.ac.uk

Bhupesh Kumar Mishra
bhupesh.mishra@hull.ac.uk

Dhavalkumar Thakker
D.Thakker@hull.ac.uk

[1] Centre of Excellence for Data Science, Artificial Intelligence and Modelling (DAIM), University of Hull, Hull, UK

[2] School of Digital and Physical Sciences, Faculty of Science and Engineering, University of Hull, Hull, UK

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	Network
GLaM	Generalist Language Model
GLUE	General Language Understanding Evaluation Benchmark
GNN	Graph Neural Networks
Google-RE	Google-Relation Extraction
GSM8K	Grade School Math 8K
JointLK	Joint Reasoning with Language Models and Knowledge Graphs
K-BERT	Knowledge-enabled Bidirectional Encoder Representations from Transformers
Kagnet	Knowledge-Aware Graph Networks
KB-ANN	Knowledge-Based Artificial Neural Network
KELP	Knowledge Graph-Enhanced Large Language Models via Path Selection
KEPLER	Knowledge Embedding and Pre-trained LanguagE Representation
KGLM	Knowledge Graph in Language Models
KiL	Knowledge-infused Learning
Lama	LAnguage Model Analysis
LINC	Logical Inference via Neurosymbolic Computation
LLM	Large Language Model
LM	Language Model
LoT	Layer-of-Thoughts
MAWPS	Math Word Problems
MetaQA	Meta Question Answering
NM	Non-monotonic
PEFT	Parameter Efficient Fine-Tuning
PL	Propositional Logic
REALM	Retrieval-Augmented Language Model Pre-Training
ReClor	Reading Comprehension Dataset Requiring Logical Reasoning
RoG	Reasoning on Graphs
SGP-Bench	Symbolic Graphics Programs Bench
SQuAD	Stanford Question Answering Dataset
T-REx	Triples Based Relationship Extraction
TC–RAG	Turing-complete Retrieval-Augmented Generation
ToM-LM	Theory of Mind Reasoning in Language Models
WebQSP	WebQuestionSP
XNLI	Cross-lingual Natural Language Inference
LLM-ARC	Large Language Model - Automated Reasoning Critic
JAKET	Joint Pre-training of Knowledge Graph and Language Understanding
KG	Knowledge Graph
FL	Fuzzy Logic
RB	Rule-Based
SLR	Systematic Literature Review
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
LTN	Logic Tensor Network
NLM	Neural Logic Machine
NTP	Neural Theorem Prover
ILP	Inductive Logic Programming
MLN	Markov Logic Network
ProbLog	Probabilistic Logic Programming
RAG	Retrieval-Augmented Generation

1 Introduction

Recently, Language Models (LMs) have gained significant attention due to their remarkable performance and potential applications across numerous fields (Wang et al., 2023). A few popular LMs include Llama3 (Grattafiori et al., 2024), GPT, Falcon (Almazrouei et al., 2023), MedPalm2 (Singhal et al., 2025), BioBERT (Lee et al., 2020), ClinicalBERT (Huang et al., 2019), BlueBERT (Peng et al., 2020), MedNLI (Herlihy & Rudinger, 2021), and Gemini (Wang & Zhao, 2023). Despite LLMs' advanced capabilities, it is evident that further improvements are necessary to address the specific requirements of various industries. LLMs are quite effective at language generation; however, ensuring transparency in responses remains a notable limitation. Transparency and trust in LLMs are undermined by hallucinations (Ahmad et al., 2023), lack of robustness (Ailem et al., 2024), trustworthiness (Lu et al., 2024), unfairness and bias (Mirzaei et al., 2024; Lissack & Meagher, 2024), security and privacy risks (Yao et al., 2024), and limited interpretability and explainability (He et al., 2025; Ma & Wolfinger, 2024). Considering these factors, expecting complete transparency in response generation solely from a transformer-based model may be unrealistic (Cambria et al., 2024). This underscores the urgency and importance of our research in this field.

Neurosymbolic AI Traditional neural networks faced challenges similar to those encountered by LLMs today (Cambria et al., 2024). In the literature, NeSy approaches have demonstrated potential in overcoming some of these challenges by combining neural learning with symbolic AI (Hamilton et al., 2022). Compared with the substantial data requirements of transformer pretraining (Hitzler et al., 2022), symbolic components can require less training data and lower computational cost in some settings (Toorajipour et al., 2021), although these advantages depend on the task, representation and reasoning procedure (Hamilton et al., 2022). In addition, symbolic components can provide explicit structure, logical constraints, and reasoning support

that may help models handle cases beyond the training distribution. However, such capabilities arise from the interaction between symbolic and neural components within the overall hybrid system (Hamilton et al., 2022; Hitzler et al., 2022).The symbolic component of NeSy AI architectures can be implemented using knowledge graphs (KGs), rule-based engines or logic-based reasoning systems. Figure 1 illustrates these paradigms, where each implementation offers distinct advantages for structured reasoning and knowledge representation to leverage both learned patterns and explicit symbolic knowledge.

Symbolic Integration with LLMs By integrating a symbolic component with LLMs, LLMs can acquire structured knowledge, which in turn can support logical reasoning and improve explainability and interpretability. However, LLMs differ fundamentally from traditional neural networks because of their large parameter scale, language modelling abilities, autoregressive token-based generation and capacity to produce context-dependent outputs. Therefore, applying existing NeSy AI frameworks to LLMs may not fully capture the unique properties and operational requirements of LLMs. These frameworks may require adaptation to accommodate the distinctive characteristics of LLMs. Accordingly, symbolic integration with LLMs can be viewed as a specialised form of NeSy AI that requires customised integration strategies, given the distinct features of LLMs. Furthermore, integrating symbolic AI poses its own challenges, including effective knowledge representation and reasoning over large-scale language contexts. This paper addresses this gap by first reviewing established NeSy AI approaches and frameworks and then proposing a roadmap

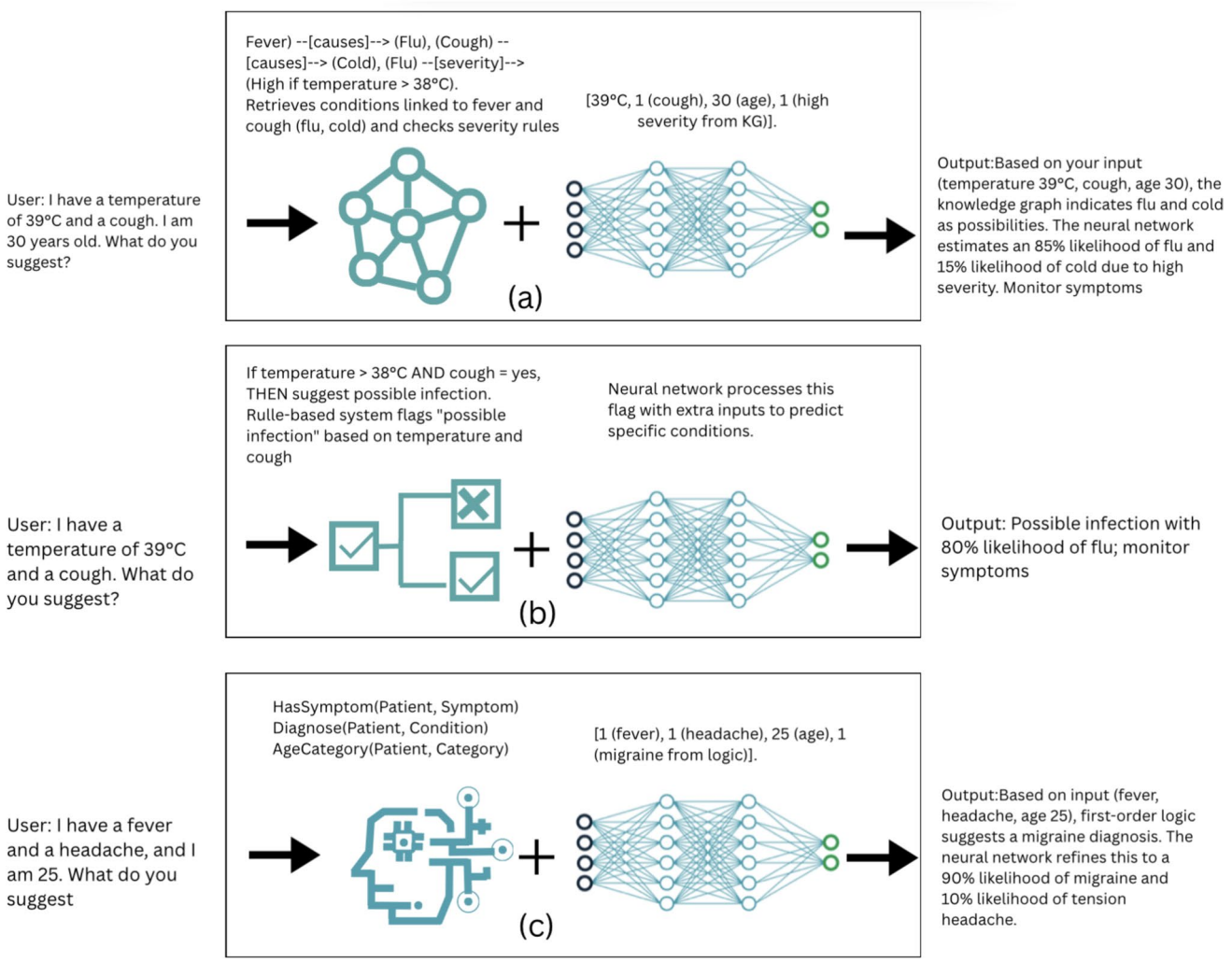


Fig. 1 Neurosymbolic AI architectures with different symbolic reasoning paradigms: (**a**) Knowledge graph-neural network integration; (**b**) Rule-Neural network integration and (**c**) Logic-Neural network integration. Each subfigure shows the information flow between symbolic and neural components with inputs and outputs at each processing stage

comprising a novel taxonomy for symbolic integration in LLMs. The main contributions of this review work include:

1. Categorisation: This article introduces a novel categorisation for combining symbolic AI with LLMs, organising approaches into categories based on integration stages in LLM development. It examines coupling levels, architectural paradigms, and frameworks at the application and algorithm levels.
2. Benchmarks: This paper critically examines the benchmarks used in the literature to evaluate methodologies that integrate symbolic AI with LLMs to improve reasoning and interpretability in LLM-generated responses.
3. Comparative analysis to address explainability: This paper examines current applications of symbolic integrated LLMs across different fields, such as reasoning, interpretability, planning, and explainability. It investigates how combining symbolic AI with LLMs can enhance explainability on numerous levels and provides a comparative analysis of existing methods, highlighting their limitations and potential future developments.
4. State-of-the-art achievements and Challenges: This article highlights state-of-the-art accomplishments and challenges in existing symbolic integrated LLMs research, with promising future research directions.

The rest of the paper is organised as follows: Section 2 presents the structured review methodology. Section 3 explains NeSy AI, including its major approaches and frameworks. Section 4.1 discusses symbolic integration with LLMs across different integration stages, including coupling mechanisms and application-level and algorithm-level perspectives. Section 5 presents the main architectural paradigms. Section 6 discusses benchmarks for KG-integrated and logic-integrated LLMs, together with quantitative benchmark analysis and associated evaluation challenges. Section 7 examines the role of symbolic integration in LLMs, including reasoning, interpretability, planning, and explainability, and provides a comparative analysis of approaches targeting explainability. Section 8 discusses state-of-the-art achievements, open challenges, and future research directions.

2 Structured Literature Review Methodology

This study employs a systematic literature review (SLR) methodology to ensure comprehensive coverage of existing literature while maintaining methodological rigour. The review was guided by the following research questions. The search terms and query construction are described afterwards:

- **RQ1:** What are the state-of-the-art methods for integrating symbolic AI with LLMs?
- **RQ2:** In what ways do these integrations enhance explainability, reasoning, and trustworthiness of LLMs?
- **RQ3:** What integration stages, coupling strategies, and architectural paradigms have been proposed?
- **RQ4:** What evaluation practices, benchmarks, and datasets are used to assess symbolic integration with LLMs?
- **RQ5:** What are the key achievements, open challenges, and future research directions in this domain?

The search strategy was developed by identifying a set of precise keywords that reflect the core dimensions of this review. These included: 'Symbolic AI', 'Large Language Models (LLM),' 'Symbolic Integration', 'Knowledge Graph (KG) Integration', 'Logic Integration', 'Explainability', 'Reasoning', 'Benchmarks', 'Challenges' and 'Achievements'. From these keywords, search components were derived to capture specific aspects of LLM–symbolic integration. The search components included 'Symbolic integration with LLM', 'KG OR logic integration with LLM', 'Symbolic integration with LLM for explainability', 'KG OR logic integration AND explainability', 'Symbolic reasoning AND LLM', 'KG integration in the pre-training stage', 'KG integration in the fine-tuning stage', and 'KG integration in the post-training stage'. Together, these combinations captured integration stages, coupling strategies, algorithm-level and application-level methods, architectural paradigms, benchmarks, roles, achievements and challenges. To ensure precision and reduce irrelevant retrievals, the search queries were adapted to the indexing requirements of each database. Eligibility of studies was determined by applying inclusion and exclusion criteria tailored to the role of symbolic integration in LLMs.

Table 1 summarises these criteria, categorised according to conceptual components of integration. This review considered studies published between 2018 and February 2025. The relatively low number of studies from early 2025 is attributed to the limited availability of articles at the time of data collection. This study employs the PRISMA 2020 guidelines to conduct a systematic literature review (SLR) (Page et al., 2021). Extensive searches were conducted across Scopus, IEEE Xplore, ACM Digital Library, SpringerLink, Elsevier (ScienceDirect), PubMed, MDPI, ACL Anthology, AAAI, IJCAI, OpenReview, and arXiv. ArXiv was used as a supplementary source to capture recent developments in this rapidly evolving area that may not yet have appeared in indexed conference proceedings or journal

Table 1 Search keywords along with Inclusion and Exclusion criteria

Conceptual Category	Inclusion Criteria	Exclusion Criteria
Symbolic AI and LLMs	Integration of Symbolic AI with LLMs to address challenges of LLM	Focusing solely on traditional neural networks without LLM components
NeSy approaches	NeSy AI frameworks and approaches addressing neural challenges	NeSy AI frameworks and approaches focusing on symbolic AI
Symbolic AI integration at several stages of LLM	Symbolic AI integration at various stages of LLM including pre-training, inference, post-training, fine-tuning	Studies not explicitly specifying the integration stage of LLM or lacking methodologies for such integration
NeSy coupling	Symbolic integration interaction level/ tight /loose coupling	Integration with LLMs at several coupling levels
Algorithm level symbolic integration with LLMs	Symbolic logic and KG-based integration to enhance LLM responses at algorithm level	Other symbolic approaches at the algorithm level that do not use symbolic logic or KG-based integration to enhance LLM responses
Application level	Symbolic logic and KG-based integration to enhance LLM responses at application level	Other symbolic approaches at the application level that do not use symbolic logic or KG-based integration to enhance LLM responses
LLM-to-Symbolic flow	Focused only on information flow from logic or formal methods conversion, action schema generation to enhance LLMs	Studies that do not involve LLM-generated data feeding into symbolic systems or lack methodological details on this integration flow
Symbolic-to-LLM	Papers demonstrating enhancements in explainability, reasoning, or interpretability through symbolic AI	Research lacking information flow direction or specific integration methodologies
Symbolic integration with LLMs Hybrid Models	Studies focusing on hybrid models combining symbolic AI and LLMs to enhance reasoning, interpretability, or task performance	Studies that do not explicitly address hybrid architectures or lack detailed integration frameworks of symbolic AI with LLMs
Symbolic Integration Benchmarks	Knowledge Graphs (KGs) and Symbolic Logic Benchmarks to address interpretability and reasoning	Studies that do not provide or utilize specific benchmarks, datasets, or evaluation metrics for symbolic integration with LLMs are excluded
The role of symbolic AI integration	Studies examining the role of symbolic AI integration in enhancing LLM capabilities such as reasoning, explainability	Studies that do not address the impact or role of symbolic AI in LLM functionality or lack practical insights into its applications
State-of-the-art challenges and achievements	Studies highlighting state-of-the-art challenges and achievements in symbolic AI integration with LLMs	Studies that do not address current challenges or notable advancements in symbolic AI and LLM integration

databases. Because arXiv hosts preprints that may not have undergone peer review, such records were screened with additional caution and were included only when they satisfied at least two of the following indicators: (i) publication in 2023, 2024, or early 2025 (up to February 2025), (ii) citation count greater than 30, and (iii) high topical relevance to the review questions. These criteria were used as pragmatic screening indicators to improve coverage of emerging work. In addition, Google Scholar was used as a supplementary source to capture grey literature and cross-check coverage. Records from Google Scholar were subsequently mapped to the corresponding databases to avoid duplication.

The initial database search returned a total of 564 records. After removing 127 duplicates, 437 records remained for screening. Titles and abstracts were screened to exclude 183 clearly irrelevant papers. The full texts of 254 reports were sought. Five reports were unavailable, leaving 249 reports for full-text eligibility assessment. During this stage, Seventy-two reports were excluded: 12 were conceptual discussions or opinion pieces, 27 fell outside the scope of symbolic integration with LLMs, and 33 provided insufficient methodological detail. Finally, 177 studies were included in the review for qualitative synthesis. These studies form the evidence for the analysis presented in this paper. The venue analysis indicates that ACL Anthology contributed the most significant number of publications. Year-wise trends highlight a sharp rise in publications from 2019 onwards, peaking in 2023–2024. The study identification, screening, and inclusion process is summarised in the PRISMA 2020 flow diagram (Fig. 2) (Page et al., 2021). The distribution of the included studies across publication years and venues is shown in Fig. 3a and b. In addition to the temporal and database coverage (Fig. 3a–b), the venue quality overview (Fig. 3c–d) indicates that journal publications predominantly appear in Q1 outlets, with a smaller share in Q2 and negligible Q3/Q4 presence. The conference set is similarly concentrated in flagship A/A venues*, with modest representation in B-tier and limited contributions from workshops/other tracks. This distribution reflects an emphasis on high-quality venues across both journals and conferences.

For each study, we extracted details on the integration method, symbolic component, coupling degree, application vs algorithmic analysis, architectural paradigms, integration role, application domain, transparency aspects, and

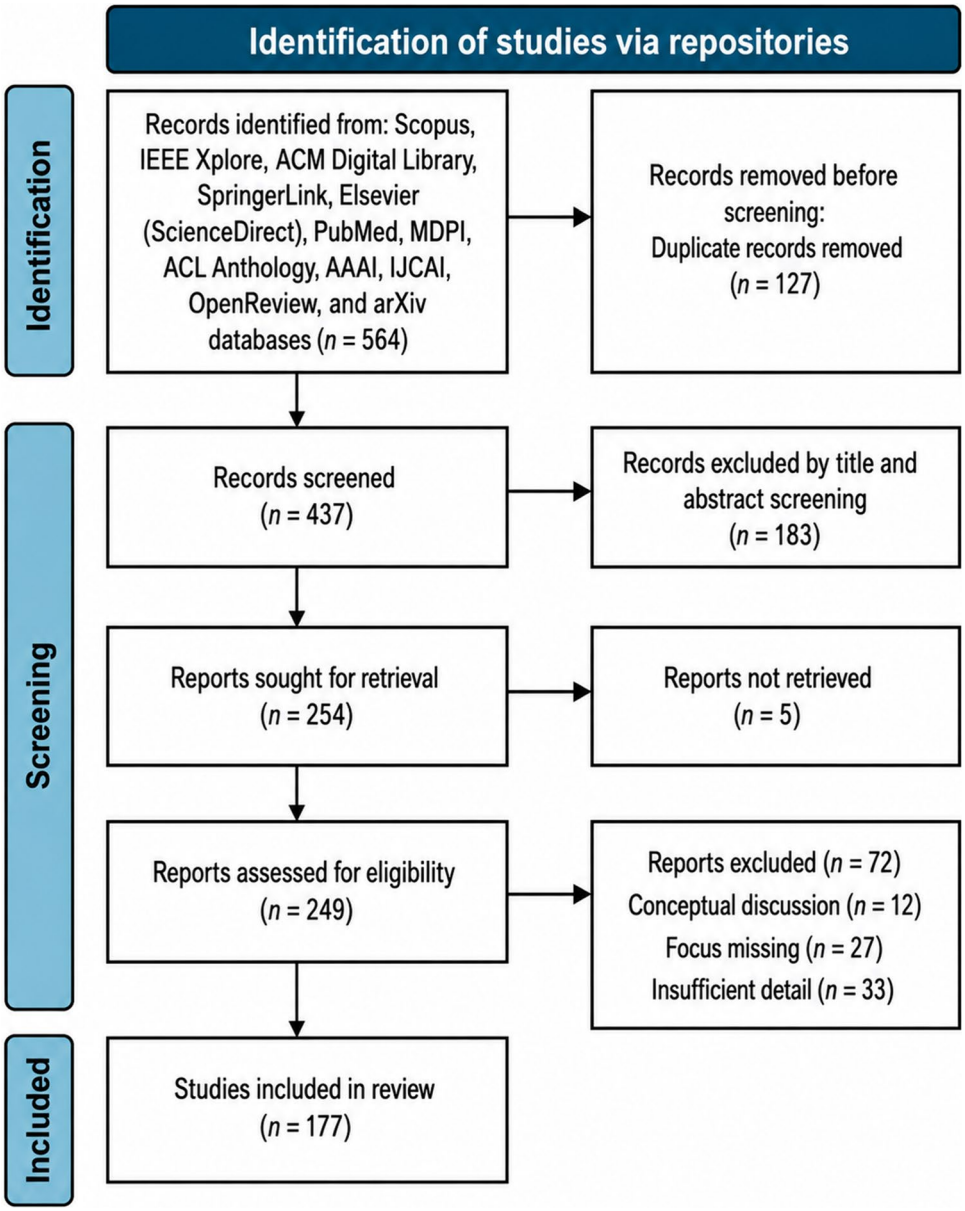
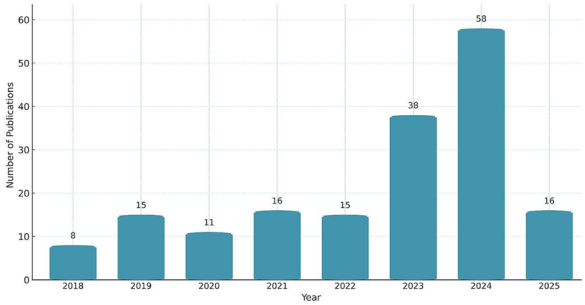
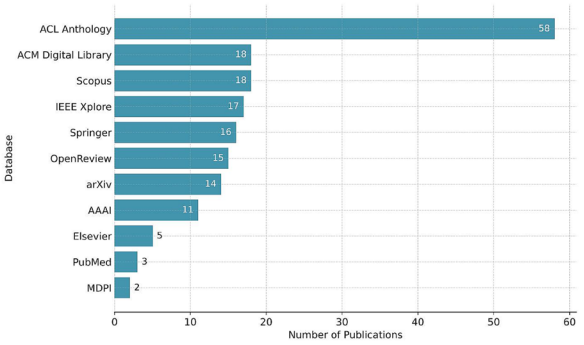


Fig. 2 PRISMA flow diagram showing the systematic literature review process (Page et al., 2021)

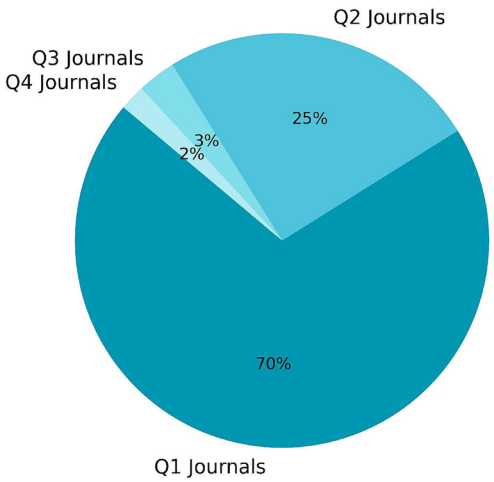
identified challenges. A narrative synthesis was conducted to systematically group and classify the included literature, enabling the identification of thematic patterns, methodological approaches, and conceptual frameworks across studies. A novel taxonomy spanning four dimensions is designed, including integration stages, coupling strategies, algorithmic versus application-level approaches and architectural paradigms. A range of existing surveys examine symbolic integration in neural approaches, but without a sustained focus on LLM-centred integration. One recent survey reviews neuro-symbolic AI over the past two decades, drawing on a broad literature base, including books, and discussing themes such as learning, reasoning, scalability, explainability, decision-making, and ethics (Bhuyan et al., 2024). However, its emphasis remains largely on neuro-symbolic AI in the broader sense, rather than on application-level or tool-level integration involving LLMs. Another systematic review examines symbolic knowledge extraction and injection in neural approaches, but gives only limited attention to integration stages (Ciatto et al., 2024). More recent concurrent surveys have considered symbolic integration with LLMs from complementary perspectives. Yang et al. (2025) focus specifically on improving the reasoning abilities of LLMs through neuro-symbolic approaches, whereas Delvecchio et al. (2025) adopt a task-directed organisation of neuro-symbolic methods in the black-box model era. This review differs in its organising principle: rather than grouping prior work by task type or reasoning capability, it develops a structural design space showing where symbolic information enters an LLM pipeline and how the symbolic



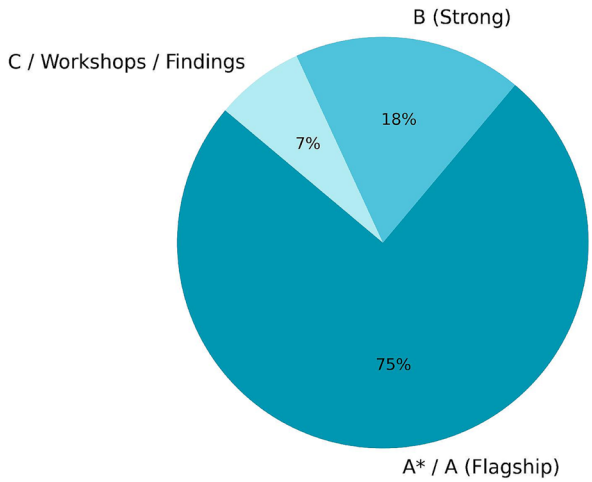
(a) Temporal distribution of studies by year



(b) Database distribution of selected studies



(c) Journal quartiles



(d) Conference tiers

Fig. 3 Distribution analysis of included studies in the systematic literature review: (**a**) Temporal spread by year, (**b**) Database coverage, (**c**) Journal quality distribution across quartiles, and (**d**) Conference quality distribution across tier groups

and neural components interact. This provides a systematic basis for analysing symbolic integration in LLMs and for structuring the subsequent discussion of challenges and future research directions.

The following overview clarifies how the remainder of the paper is organised. Figure 4 provides a structured overview of the main sections of this review. It shows how the discussion develops across different perspectives on symbolic integration with LLMs and serves as a roadmap for the sections that follow.

3 Neurosymbolic AI

3.1 Approaches in Neurosymbolic AI

NeSy AI integrates neural architectures with symbolic components. These components may include knowledge graphs (KGs) and logic-based reasoning systems, including First-order logic (FOL), Fuzzy logic (FL), Propositional logic (PL), and Rule-based systems (RB).

- FOL is a formal system that represents objects, their properties, and relationships in quantified variables, such as “for all” and “there exists”, to refer to specific elements within a domain.
- FL is used for reasoning under uncertainty, utilising truth values that range between 0 and 1, rather than being limited to the binary values of true or false.
- PL is a basic logical system which represents statements as true or false. To form more complex expressions, these statements can be combined using logical connectives like “and” “or,” and “not.”
- RB systems are AI systems that rely on predefined “if-then” rules to infer conclusions or trigger actions.

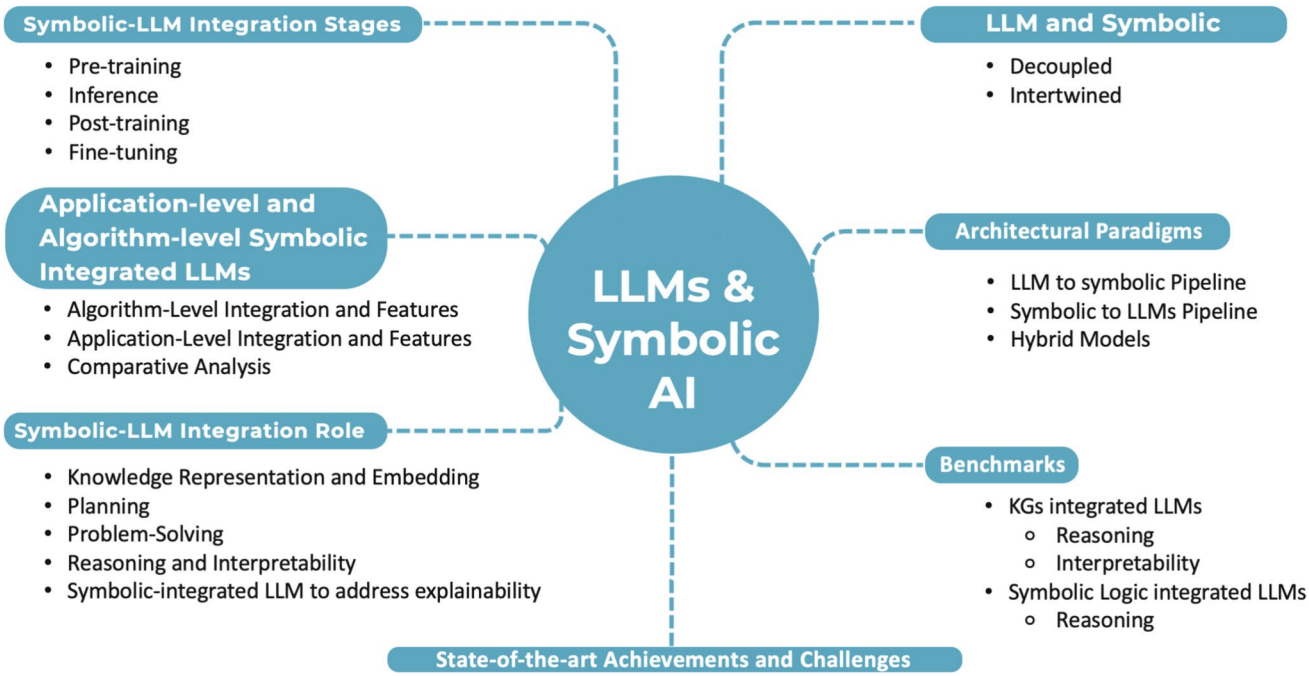


Fig. 4 Roadmap for symbolic integration with Large Language Models (LLMs)

NeSy approaches are typically classified based on the interaction between neural networks and symbolic reasoning. These include sequential, nested, cooperative, and compiled models, each offering different ways to integrate learning and reasoning components (Hamilton et al., 2022), presented in Fig. 5. Sequential is the standard deep learning approach, where neural and symbolic components perform consecutively in a series. In nested approaches, the symbolic solver performs logical tasks using a deep learning approach. In cooperative approaches, the neural component interacts with the symbolic component to make certain decisions. In compiled approaches, symbolic logic is incorporated into the training set of the neural model. In tightly coupled systems, both symbolic and neural components are integrated tightly to coordinate performance. From the perspectives of reasoning and learning, NeSy systems are classified into two broad categories: (i) high reasoning and low learning, and (ii) low reasoning and high learning.

High Reasoning, Low Learning In high-reasoning and low learning systems, the neural component plays a limited role,

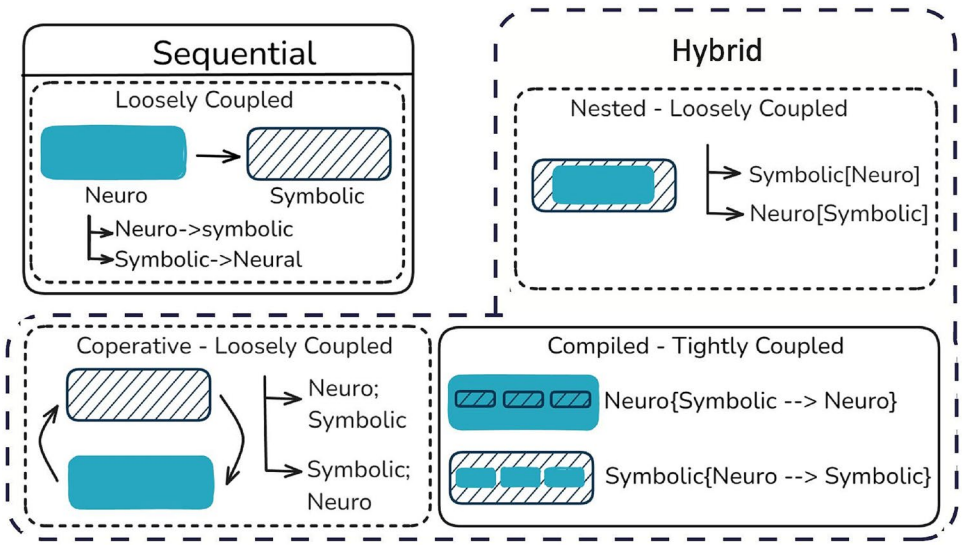


Fig. 5 Neurosymbolic approaches depending on the interaction level

Table 2 Neurosymbolic categories and approaches

Category	Approaches	Description	Example Tasks
High reasoning and low learning systems	Probabilistic Logic Programming (ProbLog)	A probabilistic programming language to define probability distribution and truth values generation by reasoning mechanism (Dries et al., 2015).	More focused on machine learning; can perform marginal inference; capable of modelling and reasoning about uncertain domains (Manhaeve et al., 2021).
	Markov Logic Network (MLN)	Applies the idea of Markov networks to logical problems and can generate inferences by learning logical rules (Marra & Kuželka, 2021).	–
	Inductive Logic Programming (ILP)	A classical rule-based approach using logic programming to represent knowledge.	Capable of providing self-explanatory symbolic, goal-driven, and causal interpretation of data (Zhang et al., 2024)
Low reasoning and high learning systems	Regularisation models	Symbols are added to objective function during training.	Can prevent overfitting by handling noisy data.
	Knowledge transfer models	Knowledge Graphs (KGs) are integrated with neural architectures.	Zero-shot learning and few-shot learning.

Table 3 Architecture of popular frameworks for neurosymbolic AI

Framework	Architecture	Syntax	Learning & Inference	Tasks	Paradigms
Logic Tensor Network (LTN)	Symbolic + Neural	First-order logic	Parameterised; Model	Semi-supervised; Distantly supervised	Logic; Neural
Neural Logic Machine (NLM)	Symbolic	Relational	Parameterised; Structured; Proof	KG completion; Knowledge induction	Logic; Neural
Neural Theorem Prover (NTP)	Symbolic + Neural	Relational	Parameterised; Structured; Proof	KG completion; Knowledge induction	Logic; Neural
DeepProbLog	Symbolic + Neural	First-order logic	Parameterised; Structured; Model; Proof	Distantly supervised; Knowledge induction	Logic; Probability; Neural

while symbolic representations and reasoning make the main contribution to prediction (Manhaeve et al., 2021).

Low Reasoning, High Learning Low-reasoning and high-learning systems use neural networks to make predictions, and symbolic knowledge or constraints make a comparatively limited contribution during training (Badreddine et al., 2022). Different approaches to performing various tasks for high-reasoning, low-learning, and low-reasoning, high-learning systems are mentioned in Table 2.

3.2 Frameworks for Neurosymbolic AI

A framework is a structured approach that provides essential components and guidelines for developing algorithms (Ma & Wolfinger, 2024). Popular frameworks for NeSy AI include Logical Tensor Networks (LTN) (Badreddine et al., 2022), DeepProbLog (Manhaeve et al., 2021), Neural Logic Machines (NLM) (Dong et al., 2019), and Neural Theorem Prover (NTP) (Minervini et al., 2018).

Logical Tensor Networks (LTN) LTNs integrate logic with neural networks by representing logical formulas as tensors and applying them within a neural network. This approach enables reasoning and learning over structured data, leading to improved inference.

DeepProbLog DeepProbLog enhances the probabilistic logic programming system by incorporating deep neural networks. This framework enables the system to learn probabilistic models directly from data, amalgamating the expressive capabilities of logic programming with the adaptability of deep learning. It supports tasks that require both logical reasoning and uncertainty modelling.

Neural Logic Machines (NLM) NLMs combine neural networks with symbolic logic using a differentiable logic engine. This enables the model to perform logical learning and reasoning tasks using tensors activated by logical rules, such as deductive, inductive and abductive reasoning.

Neural Theorem Prover (NTP) NTP integrates symbolic reasoning with deep learning methodologies to analyse mathematical theorems. It establishes their validity by identifying patterns and generalisations from extensive datasets comprising known theorems and proofs. This neural network-based automated theorem-proving strategy employs Prolog

syntax to conduct logical reasoning by embedding logic within neural networks.

These frameworks combine symbolic reasoning with neural learning, enabling machines to process structured logical rules in a form suitable for neural computation. In doing so, they support generalisation from specific cases while bridging traditional logic-based AI with data-driven neural approaches. Table 3 summarises the main architectural characteristics of these frameworks, including their syntax, inference style, learning mechanisms, tasks, and paradigms. In particular, the frameworks differ in whether they use propositional, first-order, or relational logic, and whether they rely on probabilistic or fuzzy semantics to handle uncertainty. The table also distinguishes proof-based and model-based inference, thereby highlighting how different frameworks support knowledge representation, induction, and generative reasoning.

3.3 From Neurosymbolic AI to Symbolic-Integrated LLMs

Why Conventional NeSy AI is not Sufficient? NeSy AI approaches, such as high-reasoning/low-learning systems and low-reasoning/high-learning systems, were designed for traditional neural networks and have proven effective (Hammond & Leake, 2023). In such approaches, symbolic and neural components are coupled through joint training or differentiable logic layers. NeSy frameworks like DeepProbLog, Logical Tensor Networks, NLM, and NTP were developed for conventional neural architectures and can support compositional reasoning (Bizzarri et al., 2025). However, large-scale real-world deployments remain comparatively limited. Considering the strengths and weaknesses of symbolic systems and LLMs as illustrated in Fig. 6, this integration can result in more practical and trustworthy solutions (Dinu et al., 2024). However, integrating LLMs with NeSy approaches and frameworks to enhance LLM capabilities is an early research stage, with some existing approaches pursuing theoretical perspectives. A notable theory-focused study by A. Sheth et al. presents a theoretical framework for NeSy integration with LLMs to enhance algorithmic capabilities such as reasoning and application-level features such as explainability (Sheth et al., 2023). Apart from theory-focused approaches, some approaches target application-level LLM-based NeSy integration. However, these approaches mainly use an LLM to improve the symbolic component rather than using symbolic methods to improve the LLM (Fang et al., 2024). One strategy that utilises LLMs for automatically extracting features to enhance symbolic parts is outlined in (Al-Negheimish et al., 2023), which uses LLM-generated features to strengthen the symbolic component rather than integrating symbolic elements to enhance the LLM.

Why Symbolic-Integrated LLMs Require a Distinct Perspective? LLMs are large-scale neural networks, typically based on transformer architectures, and differ operationally from the conventional neural models for which many NeSy frameworks were designed and often integrate a separately maintained symbolic component rather than jointly training both components end to end. The scale and pre-training/fine-tuning workflow of LLMs can make end-to-end symbolic coupling expensive or impractical, although objective-level and representation-level coupling remain possible. In practice, symbolic integration with LLMs is typically performed by fine-tuning, prompt engineering, or external knowledge injection, rather than through joint-training strategies. Therefore, NeSy general frameworks are not directly

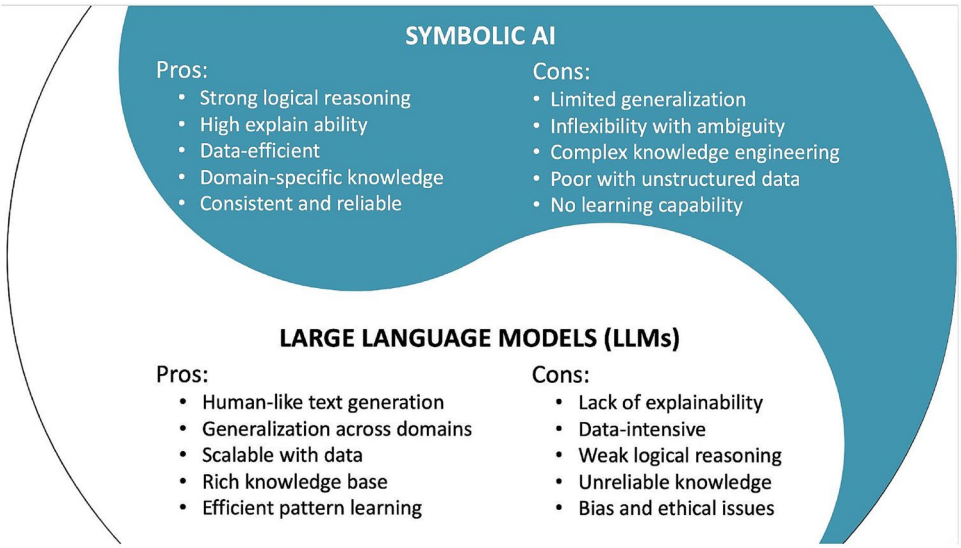


Fig. 6 Strengths and weaknesses of symbolic AI and LLMs

transferable to LLMs, and symbolic-integrated LLMs can be viewed as a specialised branch of NeSy AI. This specialised branch therefore requires integration strategies tailored to LLM scale, generation and deployment workflows. This review targets existing literature for NeSy AI and then explores the integration of symbolic AI with LLMs, considering the need for improved transparency from symbolic AI.

To maintain conceptual clarity, this review uses *Neurosymbolic AI* as the umbrella term for approaches that combine neural and symbolic methods. The expression *symbolic-integrated LLMs* is used more specifically to denote the LLM-focused branch examined in the later sections. The abbreviation *NeSy* is used only where brevity is needed. Table 4 presents a comparative analysis of various symbolic-integrated LLMs. By examining the performance and limitations of each approach, the study highlights how different integration strategies affect LLMs' capability and scalability. The focus is on evaluating the strengths and weaknesses of each method in terms of reasoning, explainability, flexibility, and computational efficiency. This comparison offers insight into the current state of NeSy integration and identifies areas for improvement to optimise LLMs for more complex, interpretable, and robust AI systems.

4 Symbolic-Integrated LLMs

4.1 Integration Stages and Mechanisms

Integrating LLMs with symbolic approaches, specifically KGs, is a rapidly evolving domain. Other than KGs, symbolic approaches include rules-based systems, logic programming, and theorem provers. This review prioritises KG-based and logic-based integration because these are the most prominent categories in the retrieved literature and

Table 4 Comparative analysis of symbolic-integrated LLM approaches

Study	Task	Neural Part	Symbolic Part	Dataset	Performance	Limitation
(Olausson et al., 2023)	Logical reasoning in natural language	StarCoder+, GPT-3.5, GPT-4	FOL and deductive inferring	FOLIO and ProofWriter	26% higher accuracy with GPT-4 than CoT on ProofWriter (FOLIO)	Only one aspect of logical reasoning is focused, Scalability, computational costs, Better reasoning and logic techniques can be used
(Gao et al., 2023)	Mathematical Reasoning and symbolic interpreter	CODEX, COT (Minerva, PaLM)	Arithmetic and symbolic reasoning	GSM8K, SVAMP, ASDIV, MAWPS, BIG-Bench	GSM8K outperforms (COT by absolute 15% top- 1 accuracy.)	
(Al-Negheimish et al., 2023)	Reasoning in Machine Reading Comprehension	GPT3.5 Turbo, LLaMa 7B, Alpaca 7B	Numerical reasoning	Manually created dataset	Improved performance for all LLMs	Extend the task to allow for nested reasoning and learning commonsense knowledge to solve complex questions beyond numerical reasoning
(Grand 2023)	Translating language into code to solve programming tasks	Codex gpt-3.5-turbo-0301, and GPT-4	Symbolic compression module	REGEX, CLEVR, LOGO	LOGO 32.13% , CLEVR 88.67% , REGEX 76%	Search does not condition on Input output examples, instead it focuses on patterns in description solution pairs
(Dong et al., 2023)	Storytelling	OpenAI's Code-LLM(GPT3), Codex	Reason	Prompt	ROC-AUC score CoRRPUS (specific functions) 79%	CoRRPUS produced code is not guaranteed to run and completely accurate in reasoning
(Fang et al.,2024)	Automated agents and Symbolic Reasoning in Games	OpenAI LLMs	Rule based reasoning for math, map reading, common sense, sorting		88%	More explained prompts can enhance LLM agent actions, symbolic modules can be added to handle complex scenarios
(He-Yueya et al.,2023)	Math word problems solver	Codex	External symbolic solver to solve equations	GSM8K math problems, new dataset ALGEBRA	Outperforms (PAL) by 20% (on ALGEBRA)	Declarative prompts with SymPy, a Python library for symbolic computation
(Enström and Kjellberg 2023)	General	BART	PL, Inference as forward and backward chaining	SimpleLogic dataset (PL)	Label-Priority, Rule-Priority, Balanced Rule priority	Bottom-up and top-down approaches are not fully comparable.
(Zhang et al., 2023)	Physics	Open AI's codex model	linguistic reasoning benchmark	Manually created dataset	$R^2 = 0.759$, $p < 0.0001$ Correlations with human judgements	Integration with pragmatic interpretation, inference, and perception, to construct better representations from perceptual inputs can enhance results

provide structured knowledge or constraints that can complement probabilistic generation. In contrast, rule-based systems and theorem provers are often rigid, computationally intensive, and less adaptable to unstructured data. This section discusses KGs and logic-based symbolic integration with LLMs. Each approach's integration is explored at several stages, including *pre-training, inference, and post-training or fine-tuning*. Figure 7 illustrates symbolic components like knowledge graphs and logic integration across these stages. This stage-specific integration architecture offers unique benefits and challenges. It concludes by reviewing the state-of-the-art advancements in both techniques and highlighting the challenges associated with each method.

4.1.1 KG and Logic Integration Across Stages

KG Integration Across Stages KG integration at the pre-training stage augments the training data or objective with structured knowledge, which can improve the modelling of entities and relations (Kau et al., 2024). KG can expand input structures, convert textual inputs into trees, enrich input information using text embeddings, and optimise Word Masks (Sap et al., 2019; Liu et al., 2020). KG integration in the *training stage* of LLM modifies different transformer components to improve contextual understanding, including the self-attention mechanism. Various approaches include knowledge-infused self-attention transformers (Roy et al., 2023), focus fusion attention mechanisms (Ma et al., 2024), and dual-interleaved attention (Zhang et al., 2024). During the training stage, models incorporate additional adapters (Wang et al., 2021) and specific encoders for KG (Zhang et al., 2019) or insert additional encoding layers (Peters et al., 2019) to perform reasoning and enhance accuracy. KG integration in *inference stage*, using Retrieval Augmented Generation (RAG) provides information and context to improve factual accuracy (Zhu et al., 2024). At inference or post-processing time, KG-based modules can validate generated content using ConceptNet (Lin et al., 2019) construct knowledge-grounded prompts (Bian et al., 2021). Another complete end-to-end LLM-symbolic planner without expert intervention integrates symbolic AI into the *inference stage* of LLM (Huang et al., 2025).

Logic Integration Across Stages In the *pre-training stage*, logic rules are infused with training data for structured learning patterns to improve understanding of logic and reasoning (Pan et al., 2023). This stage includes approaches like embedding logical constructs (Wu et al., 2024) and logic-guided augmentation of symbolic data into models (Yang et al., 2024) vocabulary and pre-training corpus. At the logic-guided *fine-tuning stage*, the model is adapted for logical reasoning without changing its core architecture. Symbolic logic can be integrated into the *inference stage* using external symbolic solvers in the inference pipeline for symbolic reasoning (Pan et al., 2023).

4.1.2 Mechanisms of Symbolic Integration

The reviewed studies also exhibit three recurring analytical mechanisms through which symbolic structure influences LLM-based systems: objective-level integration, representation-level integration, and inference-level integration. These mechanisms provide a basis for interpreting how symbolic information affects internal computation and output generation across integration stages.

Objective-Level Integration Objective-level integration refers to cases where symbolic information modifies the training objective, thereby influencing learning through parameter updates during training. For example, KEPLER (Wang et al., 2021) jointly optimises masked language modelling with a KG embedding objective:

$$\mathcal{L}_{\text{KEPLER}} = \mathcal{L}_{\text{MLM}} + \mathcal{L}_{\text{KE}} \tag{1}$$

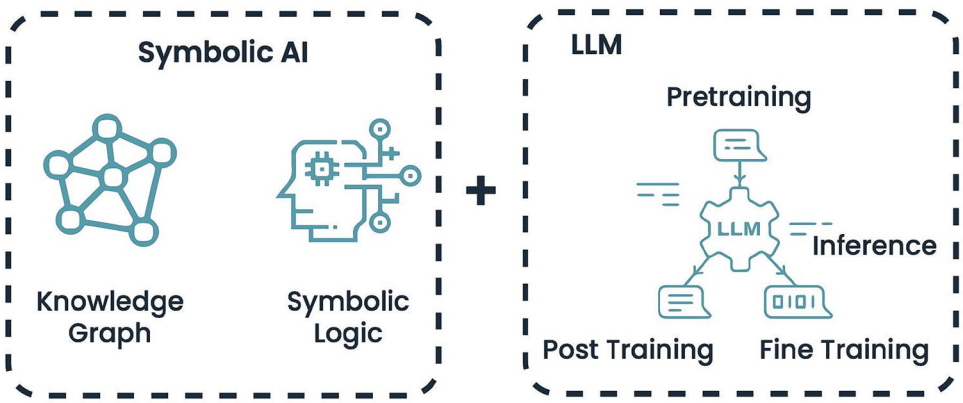


Fig. 7 Symbolic-Integrated LLM

where $\mathcal{L}_{\mathrm{KE}}$ aligns language representations with KG relational structure through knowledge embedding over positive and negative triples. This illustrates a relatively tight form of symbolic influence during learning, because symbolic structure directly affects optimisation rather than acting only through external inference-time guidance.

Representation-Level Integration Representation-level integration refers to cases where symbolic structure affects embeddings, attention, adapters, encoders, or other hidden states. Inference-level integration refers to cases where symbolic modules guide, constrain, re-rank, retrieve, or verify outputs without modifying model parameters.

4.2 LLMs and Symbolic Coupling

Coupling refers to the degree of interaction between LLMs and symbolic components. It determines how tightly both parts exchange information and how strongly symbolic constraints influence reasoning, learning, and explainability. Coupling offers solutions for various LLM challenges but introduces further challenges that must be addressed. These challenges include design complexity and balancing LLMs' flexibility with the rigidity of symbolic approaches (Sheth et al., 2023). Coupling categories based on interaction level include Decoupled integration and Intertwined integration.

Decoupled Integration In Decoupled integration, LLM and symbolic components act as autonomous elements that function independently while interacting when necessary. The autonomous structure promotes flexibility and modular design and enhances scalability. This integration does not overcomplicate the system's design and implementation, but can limit the depth of integration. The independence can lead to limited interaction and suboptimal performance, resulting in constrained contextual understanding (Sheth et al., 2023). Decoupled integration can be loosely coupled with little interaction or moderately coupled by sharing data or functionality while maintaining independence.

Intertwined Integration In Intertwined integration, LLM and symbolic components are seamlessly and tightly coupled. Such integration can be thoroughly intertwined without any independence or functionally intertwined while maintaining little independence. Loss-level symbolic integration is one of the tightest coupling approach (Wang et al., 2021). Tightly coupled approaches make the system capable of performing complex reasoning by drawing on complementary strengths of both components, resulting in a complex architecture but better overall performance. The interdependence between the systems challenges the adaptability and flexibility for modification.

4.2.1 Coupling Across Integration Stages

Symbolic coupling with LLMs can occur at different stages, including *pre-training*, *training*, *fine-tuning*, and *inference*. The degree of coupling varies according to how directly symbolic knowledge affects learned representations, optimisation, or output generation.

Pre-Training and Training *At the pre-training and training stages, symbolic knowledge is introduced to influence learned representations without being fully intertwined with the core model architecture. Coupling at these stages is therefore usually loose to moderate. DKPLM injects external knowledge triples through knowledge-enhanced pre-training (Zhang et al., 2022). KEPLER integrates KG structure directly into the training objective (Wang et al., 2021), while deterministic factual pre-training similarly incorporates structured knowledge into optimisation (Li et al., 2022). In contrast, Symbol-LLM represents a more tightly coupled approach by integrating symbolic knowledge into the injection and infusion stages of the LLM (Xu et al., 2024). Other approaches such as K-BERT (Liu et al., 2020), CoLAKE (Sun et al., 2020), and ERNIE 3.0 (Sun et al., 2021) also incorporate KG information at the pre-training stage, generally resulting in loose to moderate coupling.*

Fine-Tuning *At the fine-tuning stage, symbolic components shape task-specific adaptation without altering the model's core architecture, resulting mainly in moderate coupling. Examples include GLaM (Dernbach et al., 2024), ToM-LM (Tang & Belle, 2024), and ChatKBQA (Luo et al., 2023a). RoG (Luo et al., 2023b) also belongs to this category, although its structured reasoning paths place it between moderate and tight coupling. The full representative formulation for RoG (Luo et al., 2023b) is discussed later in Section 5.2.*

Inference *At the inference stage, symbolic components usually guide or validate generation without modifying model parameters. Coupling therefore ranges from loose to tight depending on how strongly symbolic reasoning constrains the output. Prompt- and RAG-based approaches are especially common (Pan et al., 2023), including LoT (Liu et al., 2025), CoK (Wang et al., 2024), ChatRule (Luo et al., 2023), MindMap (Wen et al., 2023), TC–RAG (Jiang et al., 2024), KGLM (Youn & Tagkopoulos, 2023), and REALM (Guu et al., 2020). More tightly coupled inference-level systems*

include LLM-Modulo (Kambhampati et al., 2024), other KG-integrated reasoning approaches (Yihao et al., 2024), semantic-parser-based integration (Dinu et al., 2024), and LAAs for LLM-empowered agents (Xiong et al., 2024).

Loss-Level Symbolic Integration A particularly tight form of coupling occurs when symbolic constraints are incorporated directly into the neural objective, allowing them to influence gradient flow and parameter updates during learning (Wang et al., 2021).

4.3 Application Level and Algorithm Level Symbolic-Integrated LLMs

This section explores the integration of LLMs with symbolic AI at the application or algorithmic level. Each approach offers unique opportunities and challenges. From application-level and algorithm-level integration perspectives, the reviewed works broadly discuss NeSy approaches that target neural architecture with symbolic data without integrating language models (Besold et al., 2017). These approaches include KB-ANN (Coatanéa et al., 2018), Logic Tensor Networks (Badreddine et al., 2022), and Logic Neural Networks (Riegel et al., 2020). However, prior surveys have less often categorised symbolic–LLM integration explicitly at the algorithm and application levels. These less prevalent approaches focus more on KG than rule-based and logic-based systems. However, several established techniques encompass both perspectives effectively, including embedding KGs with LLM using vector embeddings, adapters, or masking-based methods, which are broadly discussed in the literature (Alam et al., 2022; Xie et al., 2022; Youn & Tagkopoulos, 2023). Table 5 presents a summarised application-level and algorithm-level perspective.

4.3.1 Application-Level Integration and Features

Application-level integration refers to integrating symbolic approaches and knowledge-based systems to enhance AI capabilities in domain-specific and task-specific real-world applications. It concerns system- or workflow-level functionality designed to meet task, domain and end-user requirements, usually by connecting an LLM to external symbolic resources or tools. This includes compressing KGs, symbolic rules, or logic for integration at a particular phase to enhance LLM capabilities. Rules may be supplied by domain experts, extracted from knowledge bases or applied through rule engines and theorem provers. Logic can be first-order logic, predicate logic, or propositional logic. Logic-based LLM integration at the application level includes approaches like SymbolicAI (Dinu et al., 2024), FOLIO (Han et al., 2024), LINC (Olausson et al., 2023), and Logic LM (Pan et al., 2023). These features include multidisciplinary adaptation to domains using domain-specific knowledge such as ontologies, KGs and rule-based systems. The application level emphasises user-level features, including explainability, interpretability, safety, and trust (Sheth et al., 2023). At the application level, user explainability is a

Table 5 Application level and algorithm level perspective

Criteria	Application-Level Integration	Algorithm-Level Integration
Integration Level	Higher-level integration, typically at the system or workflow level	Lower-level integration, directly within the LLM's architecture
Primary Focus	Enhancing LLM capabilities through external symbolic components	Incorporating symbolic reasoning into the core LLM functioning
Implementation Complexity	Generally lower, as it involves combining existing tools	Higher, requires modifications to the LLM's internal structure
Scalability	Often more scalable, as components can be added or removed	Less scalable due to increased computational requirements
Explainability	Potentially higher when the external symbolic component exposes faithful, human-readable reasoning or provenance	Challenging, as integration occurs within the neural network
Performance Impact	May introduce latency because of external retrieval or symbolic processing	Faster execution, but may require more training time
Flexibility	More flexible, easier to adapt to different tasks or domains	Less flexible, changes require retraining the entire model
Coupling with LLM	Shallow, external system-level queries	Deep, integrated into training/fine-tuning
Use Cases	Task-specific enhancements, domain-specific knowledge integration	Improving general reasoning capabilities, enhancing model interpretability
Performance	Moderate, task-dependent	High, enhanced reasoning capabilities
Complexity	Low	High
Resource Requirements	Low	High
Challenges	Ensuring seamless integration, managing increased system complexity	Balancing symbolic and neural components, preserving LLM's generalization ability

crucial feature for building the confidence of stakeholders and supporting scalability for diverse use cases. Domain-specific constraints, applicability, and continual learning to ensure adaptability in diverse domains are essential features at the application level of NeSy. Application-level integration emphasises logical consistency and robustness within LLM responses by minimising errors that LLMs can generate when working independently. Real-time knowledge upgradation in LLMs using dynamic KGs or KBs is another notable feature of application-level integration. These features make application-level symbolic integration a powerful approach for enhancing LLMs' accuracy, interpretability, and reliability in real-world settings.

4.3.2 Algorithm-Level Integration and Features

Algorithm-level integration concerns computational mechanisms that embed symbolic knowledge or constraints within the model architecture, representations, objective or training process. Algorithm-level features are more tightly integrated into model functioning (Xu et al., 2024). These features are learnt and optimised during training to interact seamlessly. The seamless incorporation of features into LLM's architecture makes modification challenging. Modification may require changes to the model architecture, objective, adapters or training procedure, depending on the integration mechanism. The LLM-enhanced embedding approach in Zhang et al. (2024) is an algorithm-level integration. KiL is another algorithm-level integration containing context-adaptive algorithms to infuse knowledge into neural architectures (Gaur et al., 2022). These features include enhancing abstraction, analogy, reasoning and planning (Sheth et al., 2023). Abstraction is the ability to summarise the information by generalising it for high-level concepts. Algorithm-level integration can effectively organise knowledge hierarchically based on abstract ideas rather than concrete details. Abstraction is crucial in high-risk sectors; for instance, in medical diagnostics, an LLM could abstract specific symptoms into broader categories (e.g., "respiratory issues"), facilitating efficient diagnosis across related conditions. Analogical reasoning enables LLMs to identify structural similarities between different scenarios, using symbolic integration to map similarities across knowledge domains. This supports creative problem-solving and generalisation. Planning allows an LLM to anticipate steps needed to reach a goal, using symbolic integration to optimise decisions across multiple stages. Algorithm-level integration can enable an LLM-based system to use planning algorithms, such as Markov decision processes or symbolic task planners, to simulate outcomes and select optimal actions.

4.3.3 Comparative Analysis of Algorithm-Level and Application-Level Integration

In algorithm-level integration, features are embedded into the architecture, resulting in in-depth implementation. This contrasts with application-level features such as externally generated explanations, which are usually added as additional processes and result in shallow implementation. The operational focus at the application level is based on user interaction, whereas algorithm-level integration focuses on internal model mechanisms. Adaptability applies to the application level, whereas generalisation across various domains applies to the algorithm level. The algorithm level is less flexible, with relatively high complexity in design and implementation, but it is more efficient and cohesive. In contrast, the application level is more flexible in terms of adding new modules. Scalability depends on implementation: algorithm-level integration can provide a unified execution path but may require expensive retraining, whereas application-level integration is modular but can incur retrieval, tool and orchestration overhead. Both integrations present a spectrum of approaches, simultaneously offering significant strengths and challenges.

This section provides an overview of the integration strategies employed in combining symbolic AI and LLMs, emphasising the stages of integration, coupling mechanisms, and the application-versus algorithm-level approaches. The analysis of existing studies has identified key insights into their methodologies and limitations.

5 Architectural Paradigms

The integration of symbolic artificial intelligence with LLMs is classified into various architectural paradigms. The objective is to harness the strengths of both methodologies while alleviating their respective limitations. This section examines three primary architectural paradigms: the LLM-to-Symbolic pipeline, the Symbolic-to-LLM pipeline, and Hybrid Models.

5.1 LLM-to-Symbolic Pipeline

The LLM-to-symbolic pipeline employs LLMs to create structured symbolic representations. Symbolic engines then process these representations or knowledge structures. Various methods for converting natural language to symbolic form using LLMs have been explored in the literature to improve interpretability and reasoning. The Chain of Thought (CoT) framework is a well-known example of the LLM-to-symbolic pipeline. In CoT, the LLM produces

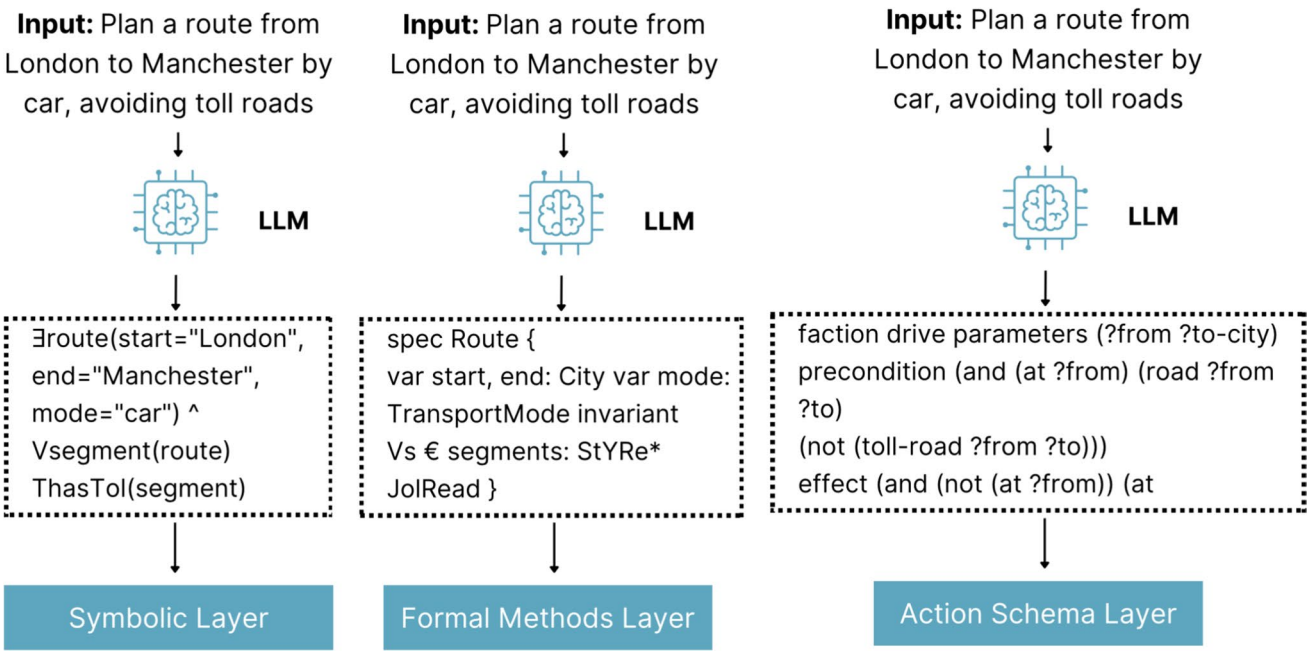


Fig. 8 LLM-to-Symbolic pipeline with three subcategories: **a**) LLM-to-Symbolic translation, **b**) LLM to formal methods translation, **c**) LLM to action schema generation

a sequence of reasoning steps in symbolic form to facilitate step-by-step reasoning. LLMs' capability to generate symbolic representations is explored in the next section. Figure 8 presents an illustration that exemplifies the pipeline from LLM-to-Symbolic representations and their respective subcategories.

Representative Formulation: Symbol-LLM To illustrate how symbolic information enters the training objective in the *LLM-to-Symbolic* pipeline, Symbol-LLM (Xu et al., 2024) provides a representative example in Eq. 2 where LLM is tuned to generate symbolic outputs directly from natural-language inputs. In the Injection stage, the model is trained to predict symbolic targets:

$$\mathcal{L}_{\text{inj}} = -\sum_{i} \log p_\theta(y_i \mid s_i \oplus x_i) \tag{2}$$

where p_θ denotes the LLM with parameters θ, x_i is the natural-language query, s_i is the instruction, y_i is the symbolic target, and $\oplus$ denotes the concatenation of the instruction with the query. In this case, symbolic structure enters the objective through supervised symbolic targets rather than through an explicit logical regulariser or constraint penalty. In the subsequent Infusion stage, symbolic data are mixed with general-language data to preserve broader language capability.

5.1.1 LLMs Generating Symbolic Representations

LLMs are deployed to create structured symbolic representations from human-understandable language to enhance the interpretability of complex concepts (Saba, 2023). When incorporated into an LLM-based system, symbolic representations can support more explicit and verifiable inference on reasoning tasks (Gaur & Saunshi, 2023). These representations may be processed through: (i) symbolic solvers or logic engines; (ii) prompts that elicit formal structures; (iii) fine-tuning on symbolic data; (iv) schema- or ontology-guided generation; (v) program-of-thought methods; (vi) syntactic or semantic parsing; (vii) iterative feedback from rule-based systems; or (viii) symbolic knowledge injection during training. LINC is a symbolically integrated LLM approach that translates problems into formal logic and uses a theorem prover to perform deductive reasoning (Olausson et al., 2023). Despite its adequate accuracy and performance, it often suffers from a syntactic and semantic mismatch between logical expression generated by the model and what the prover can process, which needs further solutions. SymbCoT uses logic rules and a chain of thoughts to translate natural language into symbolic representations to enhance reasoning abilities (Xu et al., 2024). This approach suffers from performance bottlenecks due to sequential solver operations, which demand parallel processing techniques to reduce bottlenecks.

5.1.2 Natural Language to Formal Language Translation

LLMs can automatically generate formal language specifications and proofs by translating word problems. This approach can improve challenges related to interpretability, explainability, and reasoning. nl2postcond is an approach that uses LLMs to convert language specifications into symbolic-based formal methods (Endres et al., 2024) Formal-LLM is another approach that translates natural language into formal language when designing an LLM-based agent's plan (Li et al., 2024). Another strategy for using LLMs to translate mathematical problems into formal verification properties is proposed in Riegel et al. (2020). The proposed approach employs self-prompting to perform symbolic reasoning to align with the numeric answer, thus generating an interpretable and verifiable response. Various techniques are proposed to assess LLMS' capabilities in translating formal language syntax. However, these assessment metrics are considered inadequate for determining the safety of formal translation outputs, as demonstrated by the extensive experiments in Ibrahimzada et al. (2025), highlighting a significant research gap in the literature. Furthermore, these methods have additional limitations, such as relying on static models and formalising relatively minor theories. To address this limitation, transformer memorisation (Alam et al., 2022) has been introduced, but it remains insufficient, underscoring the need for further investigation.

5.1.3 Action Schema Generation

This pipeline generates an action schema from language text to design executable plans (Kumar Jha et al., 2023). Numerous approaches used for LLM-based action schema generation include few-shot prompting (Song et al., 2023), corrective re-prompting (Raman et al., 2022), fine-tuning (Pallagani et al., 2023), agent planning (Li et al., 2024), hybrid LLM-symbolic methods (Li et al., 2024) integrating predefined logic frameworks, schema libraries (Huang et al., 2025), and automated validation modules. Methods for generating action schemas with LLMs often retain human experts in the loop, but the challenge is that human interpretations may not always align with the user's actual intent (Huang et al., 2025). To address this, the LLM-Symbolic Planning pipeline is proposed without requiring human experts in the loop (Huang et al., 2025). This pipeline designs a library of diverse action schema candidates, capturing multiple interpretations of natural language descriptions with semantic validation and ranking modules. However, such approaches remain difficult to evaluate because existing metrics may not capture the validity and executability of dynamically generated action schemas (Huang et al., 2025). The literature widely discusses LLM-generated symbolic planners (Pallagani et al., 2024), considering their capability to create intelligent planning systems. However, other work argues that LLMs should not be treated as reliable planners on their own and are better used within bidirectional "LLM-modulo" frameworks (Kambhampati et al., 2024). This dilemma underscores the need for further research and experimentation to validate the hypothesis and clarify its scope of applicability.

5.2 Symbolic-to-LLM Pipeline

The symbolic-to-LLM pipeline integrates external symbolic knowledge into LLMs, encompassing logic rules, KGs, and ontologies. This pipeline effectively empowers LLMs to produce interpretable outputs informed by structured knowledge. Numerous methodologies are explored in the literature to improve interpretability and accuracy within LLMs. A prominent example of a symbolic-to-LLM approach is the graph-based retrieval-augmented generation approach, in which relevant information from the KG

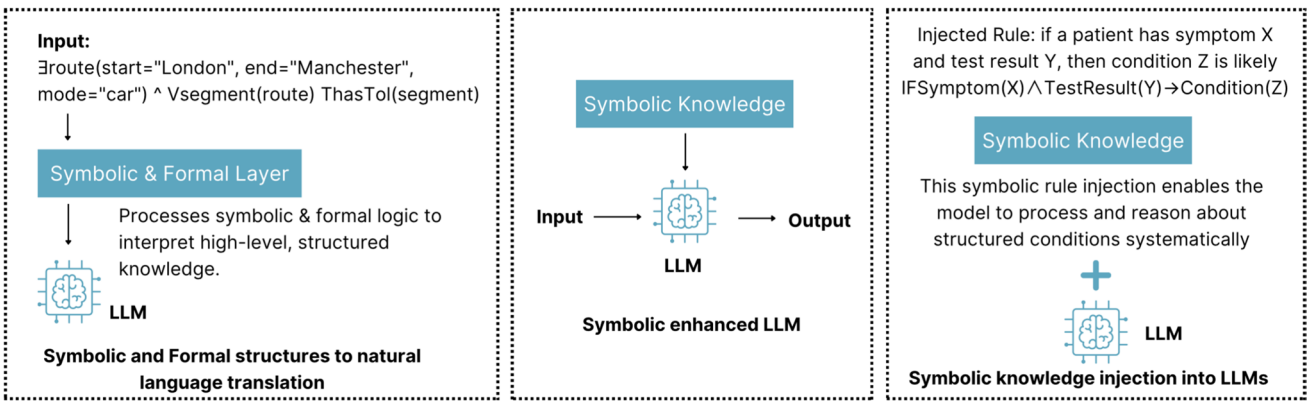


Fig. 9 Symbolic-to-LLM pipeline with three subcategories: **a**) Symbolic and formal structures to natural language translation, **b**) Symbolic enhanced LLM **c**) Symbolic knowledge injection into LLMs

is extracted and provided to the LLM to support grounded generation. Figure 9 presents an illustration that exemplifies the pipeline from Symbolic-to-LLM and its respective subcategories.

Representative formulation: RoG To illustrate how symbolic structure enters the *Symbolic-to-LLM* pipeline, RoG (Luo et al., 2023b) provides a representative example in Eq. 3 where KG-grounded relation paths act as reasoning plans. In simplified form, its objective can be written as

$$\mathcal{L}_{\mathrm{RoG}} = \log p_\theta(a \mid q, \mathcal{Z}_K^*, \mathcal{G}) + \frac{1}{|\mathcal{Z}^*|} \sum_{z \in \mathcal{Z}^*} \log p_\theta(z \mid q), \tag{3}$$

where q is the question, a is the answer, $\mathcal{G}$ is the knowledge graph, $\mathcal{Z}^*$ denotes the shortest valid relation paths used as supervision, and $\mathcal{Z}_K^*$ is a sampled subset used during reasoning. The first term represents answer generation conditioned on KG-grounded paths, while the second term supervises the generation of relation-path plans. Thus, symbolic structure enters RoG through symbolic supervision over valid relation paths and through path-conditioned answer generation.

5.2.1 Symbolic Knowledge Injection into LLMs

Symbolic knowledge incorporation into LLMs can enhance their reasoning capability and explainability. This knowledge injection can be performed at any stage, including pre-training, fine-tuning, or the inference phase. Several techniques are proposed to inject symbolic information at various stages of LLMs, including pretraining, fine-tuning, and inference. An approach incorporating structured KGs into LLMs at the fine-tuning stage is discussed in Vasisht et al. (2023), and Kagnet performs knowledge infusion into the fine-tuning stages (Lin et al., 2019). Prompting LLMs with injected knowledge at the pretraining stage is discussed in Hu et al. (2024). Injecting symbolic knowledge in training phases can cause a loss of generalisation ability in LLM (Xu et al., 2024). A two-stage tuning framework is proposed for injecting symbolic knowledge without catastrophic forgetting (Xu et al., 2024). Other knowledge injection approaches include encoding symbolic rules into input prompts, embedding logic and KGs into LLMs, ontology-based injection and prompting, constraint-based fine-tuning, Logic-Informed Fine-Tuning with Symbolic Rules, Schema and Template-Based KG Injection, and hybrid approaches for validation or refinement. These approaches improve structured reasoning capabilities in LLM. However, challenges remain, including consistency in symbolic injections with the LLM's probabilistic nature, complexity overhead and scalability. Further research into flexible hybrid architectures could help address these limitations.

5.2.2 Symbolic and Formal Structures for Natural Language Translation

Symbolic translation into language using LLMs is an evolving area of research aiming to make complex logical structures more accessible and interpretable. This process can significantly enhance the capabilities of LLMs by providing them with logically structured inputs derived from symbolic AI systems. Various approaches have been explored for translation, including symbolic input parsing by analysing components, relationships and logical connectives, predefined templates for symbolic structure conversion to natural language, and domain-specific vocabulary-based context-aware translation. Various approaches use translated symbolic content to enhance LLM outputs by providing a verifiable reference for additional context. These approaches include supplying logical explanations for complex reasoning steps and improving fact-checking processes in feedback mechanisms and reinforcement approaches, thus serving as a reliable verification layer to support LLM. S2L is an approach which converts symbol-related problems into language-based representations by prompting LLMs or leveraging external tools (Hu et al., 2024). Language-based representations are integrated into the original problem, providing in-depth contextual information to assist LLM in enhancing interpretability. Symb-XAI is another approach which converts symbolic logic to natural language to express different logical relationships between input features (Schnake et al., 2025). Another code-based symbolic translator using Reinforcement Learning with Feedback is discussed in Jana et al. (2023).

5.2.3 Symbolic-Enhanced LLM

Symbolic-enhanced LLM is another approach explored in the symbolic-to-LLM pipeline for enhancing LLM outputs by leveraging external symbolic systems. The symbolic solvers can be logic engines or KGs to supply structured, factual, or logically consistent information that the LLM doesn't inherently contain. In the KG-Enhanced LLM Reasoning framework, the KG provides curated, relational data that the LLM does not inherently possess, allowing the LLM to develop more accurate, contextually relevant, or logically sound outputs. In KG-enhanced LLMs, KG retrieves relevant information, which is then used to augment LLM prompts. A KELP framework is proposed to extract useful information from KG and send it to LLM to address the hallucination (Liu et al., 2024), contrary to approaches where LLMs construct KGs and extract information from KGs (Zhang & Soh, 2024). The symbolic system leverages symbolic solvers to perform logical reasoning on pre-defined logical statements and supplies this processed information

to the LLM. Apart from common challenges, including data sparsity, coverage, and scalability in annotated logic corpora, this area demands further research to deal with less structured or ambiguous information, such as open-ended queries.

5.3 Hybrid Models

This paradigm includes bidirectional and iterative architectures in which neural and symbolic components are integrated within a unified framework, covering both symbolic-integrated LLM architectures and modular hybrid systems. The aim is to iteratively leverage the strengths of each system to compensate for the weaknesses of the other. Bidirectional approaches facilitate continuous exchange between LLM and symbolic systems to ensure enhanced explainability. Song et al. (2023) proposes a framework for dynamic collaboration between LLM and symbolic systems to tackle problem-solving and hybrid thinking. Another approach leverages an iterative hybrid architecture to perform continuous reasoning until a successful answer is obtained (Lai et al., 2023). Despite better interpretability, such systems need help with the complexity of the interaction pipeline between the components. Neurosymbolic AI combines neural networks' learning capabilities with symbolic systems' reasoning abilities. ProRef is a symbolic-integrated LLM approach that applies various prototypes to improve logical reasoning using LLMs (Chen, 2025). Modular systems offer flexible integration, combining plug-and-play components in multiple ways depending on the task. LLM-Modulo Framework is proposed to combine the strengths of LLMs with model-based verifiers for more flexible problem and solution specifications (Kambhampati et al., 2024). Hybrid architecture needs more literature targeting seamless integration to address inconsistencies and conflicts in reasoning. The computational complexity associated with integrating symbolic reasoning and large-scale neural networks can lead to heightened latency and resource demands, thereby complicating the efficient implementation of real-time applications. Hybrid models promise tighter integration but can be challenging to design and train effectively. Modular systems offer flexibility but require careful design to ensure effective communication between components.

5.4 Consolidated Taxonomy Summary

To improve the navigability of the reviewed literature, Table 6 consolidates representative studies across the four dimensions of the proposed taxonomy. Beyond summarising prior work, the table also reveals a clear distributional pattern. Most existing approaches are concentrated around inference-stage integration, moderate or loose coupling, and application-level designs. Comparatively fewer studies target earlier intervention points such as pre-training and training, tightly integrated systems, or algorithm-level architectures. This concentration is analytically meaningful. It suggests a broader preference in the literature for integration strategies that preserve modularity, avoid deep architectural modification, and reduce engineering cost. The following subsection interprets the taxonomy as a framework for understanding how design choices alter optimisation, inference, and explainability.

5.5 Analytical Implications of the Proposed Taxonomy

This subsection interprets the proposed taxonomy as an analytical framework for understanding how symbolic structure affects optimisation, inference, system integration, and control flow in symbolic-integrated LLMs. Table 7 summarises the analytical implications of the four taxonomy dimensions and the associated recurrent formal patterns. These formulations are analytical abstractions of recurrent design patterns in the reviewed literature rather than exact descriptions of every method.

Integration Stages At pre-training and fine-tuning stages, symbolic information can influence learning objectives or task-specific adaptation, whereas inference and post-training approaches typically preserve model parameters and instead shape retrieval, verification, or reasoning control. Analytically, this distinction separates methods that alter learned representations from those that guide generation externally. This also helps explain why inference-stage integration is especially common in the literature: it is comparatively modular and easier to deploy (Dernbach et al., 2024; Guu et al., 2020; Jiang et al., 2024; Li et al., 2022; Liu et al., 2025; H. Luo et al., 2023a; L. Luo et al., 2023b, 2023; Ma et al., 2024; Peters et al., 2019; Sun et al., 2020, 2021; Tang and Belle, 2024; Wen et al., 2023; J. Wang et al., 2024; X. Wang et al., 2021; Youn and Tagkopoulos, 2023; Zhang et al., 2022; Zhu et al., 2024).

Coupling Strategies In decoupled integration, symbolic components remain external and mainly guide the LLM through prompting, retrieved context, or validation. In intertwined integration, symbolic structure affects hidden-state computation, optimisation, or inference logic more directly. In analytical terms, this creates a trade-off between modularity and flexibility on the one hand, and stronger reasoning control on the other (Jiang et al., 2024; Kambhampati et al., 2024; Luo et al., 2023; Wang et al., 2024; Wen et al., 2023;

Table 6 Consolidated summary of representative reviewed works across the proposed taxonomy of symbolic–LLM integration

Taxonomy dimension	Category	Representative references
Integration stage	Pre-training	(Sap et al., 2019; Liu et al., 2020; Wu et al., 2024; Yang et al., 2024; Zhang et al., 2022; Wang et al., 2021; Li et al., 2022; Xu et al., 2024; Sun et al., 2020, 2021; Guu et al., 2020)
	Training	(Ma et al., 2024; Peters et al., 2019; Roy et al., 2023; Wang et al., 2021; Zhang et al., 2019, 2024)
	Fine-tuning	(Baldazzi et al., 2023; Dernbach et al., 2024; Lin et al., 2019; H. Luo et al., 2023a; L. Luo et al., 2023b; Tang & Belle, 2024; Vasisht et al., 2023; Zhang et al., 2024)
	Inference	(Dinu et al., 2024; Sheth et al., 2023; Fang et al., 2024; Olausson et al., 2023; Gao et al., 2023; Al-Negheimish et al., 2023; Grand, 2023; Dong et al., 2023; He-Yueya et al., 2023; Zhu et al., 2024; Huang et al., 2025; Pan et al., 2023; Kambhampati et al., 2024; Yihao et al., 2024; Xiong et al., 2024; Liu et al., 2025; Wang et al., 2024; Luo et al., 2023; Wen et al., 2023; Jiang et al., 2024; Youn & Tagkopoulos, 2023; Kumar Jha et al., 2023; Hu et al., 2024; Liu et al., 2024; Zhang & Soh, 2024)
	Post-training / post-processing	(Bian et al., 2021; Lin et al., 2019)
Coupling	Decoupled / loose	(Sheth et al., 2023; Gao et al., 2023; Al-Negheimish et al., 2023; Grand, 2023; Dong et al., 2023; He-Yueya et al., 2023; Zhu et al., 2024; Liu et al., 2025; Wang et al., 2024; Luo et al., 2023; Wen et al., 2023; Jiang et al., 2024; Youn & Tagkopoulos, 2023; Guu et al., 2020)
	Loose / moderate	(Liu et al., 2020; Sun et al., , 2021; Wang et al., 2021; Zhang et al., 2022)
	Moderate	(Dinu et al., 2024; Fang et al., 2024; Olausson et al., 2023; Dernbach et al., 2024; Tang & Belle, 2024; Luo et al., 2023a; Zhang et al., 2024; Baldazzi et al., 2023; Kumar Jha et al., 2023)
	Moderate / tight	(Luo et al., 2023b)
	Tight / intertwined	(Kambhampati et al., 2024; Xiong et al., 2024; Xu et al., 2024; Yihao et al., 2024)
Integration level	Application-level	(Dinu et al., 2024; Sheth et al., 2023; Fang et al., 2024; Olausson et al., 2023; Gao et al., 2023; Al-Negheimish et al., 2023; Grand, 2023; Dong et al., 2023; He-Yueya et al., 2023; Pan et al., 2023; Han et al., 2024; Hu et al., 2024; Liu et al., 2024; Zhang & Soh, 2024)
	Algorithm-level	(Baldazzi et al., 2023; Coatanéa et al., 2018; Gaur et al., 2022; Kumar Jha et al., 2023; Riegel et al., 2020; Zhang et al., 2024)
	Both / mixed perspective	(Alam et al., 2022; Xie et al., 2022; Youn & Tagkopoulos, 2023)
Architectural paradigm	LLM → Symbolic: symbolic representation generation	(Olausson et al., 2023; Xu et al., 2024; Zhang & Soh, 2024)
	LLM → Symbolic: natural language to formal language translation	(Endres et al., 2024; Li et al., 2024)
	LLM → Symbolic: action schema generation	(Huang et al., 2025; Kambhampati et al., 2024; Kumar Jha et al., 2023; Li et al., 2024; Pallagani et al., 2023, 2024; Raman et al., 2022; Song et al., 2023)
	Symbolic → LLM: symbolic knowledge injection into LLMs	(Baldazzi et al., 2023; Hu et al., 2024; Lin et al., 2019; Vasisht et al., 2023; Xu et al., 2024)
	Symbolic → LLM: symbolic/formal structures to natural language translation	(Hu et al., 2024; Jana et al., 2023; Schnake et al., 2025)
	Symbolic → LLM: symbolic-enhanced LLM	(Liu et al., 2024)
	Hybrid models	(Lai et al., 2023; Song et al., 2023)

Xu et al., 2024; Xiong et al., 2024; Yang et al., 2024; Yihao et al., 2024).

Application-Level and Algorithm-Level Integration This integration reflects how deeply symbolic structure is embedded in the overall system. Application-level integration places symbolic reasoning outside the model core through workflow-level components such as retrieval, prompting, or verification. Algorithm-level integration embeds symbolic structure more directly into representations, optimisation, or architectural components. The main analytical implication is a trade-off between adaptability and internal control: application-level designs are easier to deploy and modify, whereas algorithm-level designs can provide deeper alignment between symbolic and neural reasoning at greater implementation cost (Olausson et al., 2023; Al-Negheimish et al., 2023; Yang et al., 2024; Xie et al., 2022; Xu et al., 2024; Zhang et al., 2024).

Architectural Paradigms In *LLM-to-Symbolic* pipelines, the LLM first produces a symbolic representation that is then executed, validated, or refined by a symbolic

Table 7 Analytical implications of the proposed taxonomy

Dimension	Subtype	Formal pattern	Analytical implication	Examples
Integration stages	Pre-training / fine-tuning	$\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda\mathcal{L}_{\text{sym}}$ or $p_\theta(y \mid x, \mathcal{S})$ with symbolic supervision	Symbolic structure directly influences parameter updates or task-specific adaptation, shaping internal reasoning behaviour during learning.	KEPLER (Wang et al., 2021); K-BERT (Liu et al., 2020)
Integration stages	Inference / post-training	$y^* = \arg\max_y \big(\log p_\theta(y \mid x) + \lambda C_{\text{sym}}(y, x, \mathcal{S})\big)$	Symbolic structure guides, constrains, or verifies generation without necessarily modifying the base model parameters.	RoG (Luo et al., 2023b); Graph-based retrieval / verification (Fan et al., 2024; Peng et al., 2024)
Coupling	Decoupled integration	$y = f_\theta(x, \mathcal{S}_{\text{ext}})$, $\mathcal{S}_{\text{ext}}$ = external symbolic context	LLM and symbolic components remain relatively autonomous, favouring modularity, flexibility, and scalability, but typically offering weaker internal control.	TC-RAG (Jiang et al., 2024); REALM (Guu et al., 2020)
Coupling	Intertwined integration	$h_\ell = F_\theta(h_{\ell-1}, \mathcal{S})$ or $\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda\mathcal{L}_{\text{sym}}$	Symbolic structure is tightly embedded in hidden-state computation, optimisation, or both, yielding stronger internal alignment at higher implementation complexity.	KEPLER (Wang et al., 2021); Symbol-LLM (Xu et al., 2024)
Integration level	Application-level integration	$y = f_\theta(x, r(x), v(x))$	Symbolic modules operate at the system or workflow level to enhance explainability, interpretability, robustness, and domain adaptability.	SymbolicAI (Dinu et al., 2024); LINC (Olausson et al., 2023); Logic LM (Pan et al., 2023)
Integration level	Algorithm-level integration	$h = g_\theta(x, \mathcal{S})\mathcal{L} = \mathcal{L}_{\text{LM}} + \lambda\mathcal{L}_{\text{sym}}$	Symbolic structure is embedded within the LLM architecture or training process, enabling deeper control over reasoning but reducing modular flexibility.	KiL (Gaur et al., 2022); LLM-enhanced embedding (Zhang et al., 2024)
Architectural paradigm	LLM → Symbolic	$p_\theta(s \mid x)$, where s is a path, rule, or query	The LLM generates explicit symbolic artefacts that can be executed, checked, or used as reasoning plans.	Symbol-LLM (Xu et al., 2024); LINC (Olausson et al., 2023)
Architectural paradigm	Symbolic → LLM	$p_\theta(y \mid x, \mathcal{S})$ or constrained decoding with $\mathcal{S}$	Externally supplied symbolic structure guides generation, retrieval, or verification and thereby shapes the effective inference process.	RoG (Luo et al., 2023b); KELP (Liu et al., 2024)

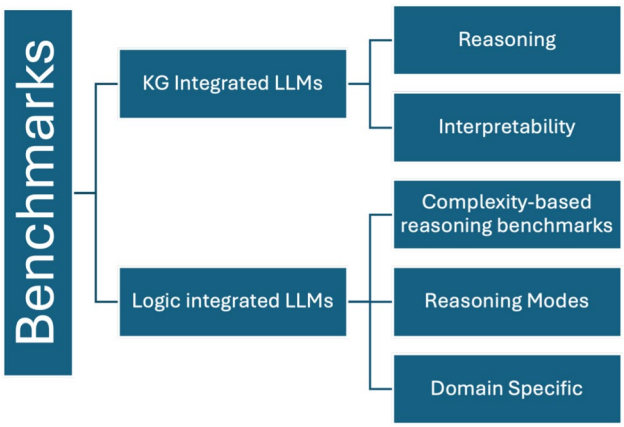


Fig. 10 Conceptual view of benchmark section

system. In *Symbolic-to-LLM* pipelines, symbolic knowledge instead serves as grounding, constraint, or guidance for generation. Hybrid models combine both directions and can support iterative refinement, although they also introduce greater coordination complexity. Analytically, these paradigms differ in whether symbolic structure is used primarily for external verification or for grounding and controlling generation (Huang et al., 2025; Xu et al., 2024, 2024; Endres et al., 2024; Li et al., 2024; Song et al., 2023; Raman et al., 2022; Pallagani et al., 2023; Vasisht et al., 2023; Hu et al., 2024; Liu et al., 2024; Zhang & Soh, 2024; Lai et al., 2023).

These distinctions show that the proposed taxonomy captures meaningful design trade-offs across symbolic-integrated LLMs.

6 Benchmarks

Benchmarks in symbolic-integrated LLMs function as standardised evaluations for assessing reasoning, explainability, and task performance. This section examines benchmarks related to knowledge graphs (KGs) and symbolic logic. The conceptual framework of the benchmark section is depicted in Fig. 10.

6.1 Benchmarks for KG-Integrated LLMs

Knowledge graphs (KGs) offer a structured representation of knowledge, making them particularly valuable for enhancing interpretability and reasoning in LLMs compared to addressing other challenges these models face. This section explores benchmark datasets in existing studies, emphasising reasoning and interpretability using KGs in LLMs. Table 8 presents benchmarks and evaluation metrics used in existing studies integrating KG with LLMs to perform reasoning and interpretability.

6.1.1 Reasoning

Integrating KGs with LLMs has created a necessity to develop benchmarks for evaluating the reasoning abilities of hybrid systems. The Chain-of-Knowledge (CoK) framework assesses and enhances knowledge reasoning in LLMs by integrating KGs (Tan et al., 2025). This framework introduces the KnowReason dataset using rule mining and transforming KG triplets into sentences. To assess the step-by-step reasoning and error identification in reasoning steps by analysing assumptions and logic in KG-integrated LLMs, the Meta-Reasoning Benchmark (MR-Ben) introduces human-curated questions across diverse domains (Zeng et al., 2024). GLUE (Wang et al., 2018) and SuperGLUE (Wang et al., 2019) are widely recognised benchmarks for KG-integrated approaches to address reasoning (Zhang et al., 2019). These benchmarks help evaluate the effectiveness of KG integration, whether used in the embedding phase or to enhance interpretability at various stages (Ibrahim et al., 2024). Benchmarks used in other approaches integrating KG for reasoning in LLMs include ATOMIC (Sap et al., 2019), XNLI (Conneau et al., 2018), ReClor (Yu et al., 2020), HellaSwag (Zellers et al., 2019), ComplexWebQuestions (Talmor & Berant, 2018), MetaQA (Zhang et al., 2017), GrailQA (Jiang & Usbeck, 2022). JointLK in Sun et al. (2022) integrates KG to perform interpretable reasoning using LM and GNN through a bidirectional attention module and evaluated on CommonsenseQA (Talmor et al., 2018). GreaseLM (Zhang et al., 2022) uses KG to reason over situational constraints and structured knowledge using three benchmarks: CommonsenseQA (Talmor et al., 2018), OpenbookQA (Mihaylov et al., 2018), and MedQA-USMLE (Jin et al., 2021). A KG-based approach discussed in Wang et al. (2020) improves reasoning in LLM on Natural Language Inference (NLI) problems for the science domain, evaluated using the SciTail science questions dataset (Khot et al., 2018). KagNet (Lin et al., 2019) is another approach using KG for commonsense reasoning on the CommonsenseQA benchmark (Talmor et al., 2018). GAIN (Zeng et al., 2020) uses double graphs with a path reasoning mechanism to provide better reasoning by interpreting relations between entities using the DocRED benchmark (Xu et al., 2024). These datasets emulate real-world situations; however, reasoning datasets suffer while analysing assumptions and scrutinising intermediate reasoning steps (Zeng et al., 2024). Ensuring consistency across reasoning paradigms and generalisation abilities also presents significant challenges (Zeng et al., 2024). Further efforts are needed to develop more advanced datasets emphasising cross-domain knowledge application, dynamic knowledge updating, and multimodal reasoning (Tan et al., 2025).

6.1.2 Interpretability

The interpretability benchmarks for KG-integrated LLMs focus on assessing the explainability of these hybrid systems. These benchmarks assess the model's reasoning process by tracing the paths to justify decisions. To assess interpretability in practical applications, the KGQA benchmark is proposed to interpret the reasoning process in LLMs enhanced with KGs for open-ended, real-world question-answering scenarios (Sui et al., 2024). LAMA (Petroni et al., 2019) is a benchmark leveraged to test the ability of LMs to retrieve relations between entities and is used for LLM interpretability discussed in Jiang et al. (2020). Another study used four benchmarks to systematically evaluate LLM interpretation with KGs in various domains (Luo et al., 2023). The benchmark uses LAMA (Petroni et al., 2019) and BioLAMA (Sung et al., 2021) to create factual questions, including two general-domain KGs: Google-RE (Petroni et al., 2019),

Table 8 Benchmarks for KG-based reasoning and interpretability

Domain	Benchmarks	Evaluation Metrics
KGs for Reasoning	GLUE (Wang et al., 2018), SuperGLUE (Wang et al., 2019), ATOMIC (Sap et al., 2019), XNLI (Conneau et al., 2018), ReClor (Yu et al., 2020), HellaSwag (Zellers et al., 2019), ComplexWebQuestions (Talmor & Berant, 2018), MetaQA (Zhang et al., 2017), GrailQA (Jiang & Usbeck, 2022), CommonsenseQA (Talmor et al., 2018), OpenBookQA (Mihaylov et al., 2018), MedQA-USMLE (Jin et al., 2021), DocRED (Xu et al., 2024), SciTail (Khot et al., 2018)	Accuracy, recall, F1 score, precision, consistency, reasoning steps, reasoning depth
KGs for Interpretability	LAMA (Petroni et al., 2019), BioLAMA (Sung et al., 2021), Google-RE (Petroni et al., 2019), T-REx (Elsahar et al., 2018), UMLS (Bodenreider, 2004), CommonsenseQA (Lin et al., 2019), SQuAD (Rajpurkar et al., 2016), WikiKG90M (Kochsiek & Gemulla, 2023), Open Graph Benchmark (OGB) (Hu et al., 2020), WebQuestionsSP (Bordes et al., 2015), FreebaseQA (Fader et al., 2014)	Completeness, recall, accuracy, fidelity, precision, speed, depth of knowledge

T-REx (Elsahar et al., 2018), and two domain-specific KGs: WikiBio (Sung et al., 2021) in the biology domain and ULMS (Bodenreider, 2004) in the medical domain. Another approach using biomedical knowledge probing benchmark named MedLAMA for interpretability (Meng et al., 2022). Furthermore, KagNet is another approach that uses a KG for commonsense reasoning, using CommonsenseQA (Lin et al., 2019). Lu et al. (2024) presents another study interpreting language models by integrating a KG using the benchmarks SQuAD (Rajpurkar et al., 2016) and Google-RE (Petroni et al., 2019). T-REx (Elsahar et al., 2018) is another benchmark used in language models to capture factual knowledge by integrating it with KGs (Li et al., 2022). Other popular benchmarks in approaches integrating KGs for interpretability in LLMs include WikiKG90M (Kochsiek & Gemulla, 2023), Open Graph Benchmark (OGB) (Hu et al., 2020) and NELL-995. Question-answer-based benchmarks include WebQuestionsSP (Bordes et al., 2015) and FreebaseQA (Fader et al., 2014). These benchmarks aim to evaluate various aspects of interpretability in KG-LLM integration, including reasoning paths, alignment between KG facts and LLM outputs, and the ability to handle complex, multi-hop queries across diverse domains.

6.2 Benchmarks for Logic-Integrated LLMs

Datasets are essential tools for assessing the capabilities of LLMs, particularly in reasoning and explainability. Researchers have developed various benchmark datasets to evaluate reasoning in symbolic integrated LLMs. This section explores multiple datasets used in symbolic integrated LLMs to highlight their role in addressing challenges in traditional LLMs, with a focus on reasoning. These datasets are categorised into complexity-level reasoning, domain-specific reasoning, and mode of reasoning. Table 9 presents benchmarks and evaluation metrics used in existing studies integrating logic with LLMs to perform reasoning.

6.2.1 Complexity-Based Reasoning Benchmarks

Reasoning datasets are categorised based on the complexity of steps designed to evaluate LLMs when integrated with symbolic AI. The benchmarks in this category focus on the level of complexity in steps while performing reasoning. These datasets assess how well symbolic AI enhances LLMs' reasoning in existing studies. This includes single-hop reasoning and multi-hop reasoning.

Single-Hop Reasoning: LLMs making direct inferences based on knowledge refer to single-hop reasoning. Various benchmarks include ProntoQA (FOL) (Saparov & He, 2022), LogicNLI (FOL), and LogicBench (PL, FOL, NM) (Parmar et al., 2024). Multi-hop reasoning with Single Logic includes datasets for assessing LLMs' reasoning by chaining through multiple steps to arrive at inference. This covers a single type of logic and only a few logical inference rules, including FOLIO (FL) (Han et al., 2024), ProofWriter (rules-based logic—FOL) (Tafjord et al., 2021), SimpleLogic (PL), and PARARULE-Plus (Clark et al., 2020)(Deductive). Existing reasoning datasets focus primarily on single-hop or multi-hop reasoning with limited inference rules. Multi-hop reasoning with non-monotonic and multiple logic covers datasets for drawing non-monotonic conclusions using multiple premises, various inference rules, and combinations. This includes multi-LogiEval (Patel et al., 2024), based on propositional, first-order, and non-monotonic reasoning. The design contains 30 inference rules and more than 60 combinations of basic inference

Table 9 Logic-Based reasoning benchmarks and evaluation metrics

Domain	Sub-domains	Benchmarks	Evaluation Metrics
Reasoning based on complexity	Single-hop reasoning	ProntoQA (FOL) (Saparov & He, 2022), LogicNLI (FOL), LogicBench (PL, FOL, NM) (Parmar et al., 2024)	Accuracy, precision, recall, F1 score, logical consistency, reasoning depth
	Multi-hop reasoning	FOLIO (FL) (Han et al., 2024), ProofWriter (Tafjord et al., 2021), PARARULE-Plus (Clark et al., 2020) (deductive), Multi-LogiEval (Patel et al., 2024)	Accuracy, consistency, reasoning depth
Reasoning modes	Deductive	bAbi-deductive (Weston et al., 2015), Winologic (He et al., 2021), WaNLI (Liu et al., 2022), BigBench-Deduction (Srivastava et al., 2023), RuleBERT-Union (Saeed et al., 2021), PrOntoQA (Saparov & He, 2022), ReClor (Yu et al., 2020), LogiQA (Liu et al., 2023)	Accuracy
	Inductive	CLUTTR-Systematic (Sinha et al., 2019)	Accuracy
	Abductive	LogiGLUE (Luo et al., 2023), AbductionRule (Young et al., 2021)	Accuracy, F1 score
Domain specific	Mathematics and physics	GSM Symbolic (Mirzadeh, 2025), MAWPS (Bhuyan et al., 2024), ARB (Sawada et al., 2023), Linguistic Physical Reasoning Benchmark (Zhang et al., 2023)	Accuracy
	Coding and computation	Abstraction and Reasoning Corpus (ARC) (Alford, 2021), CodeXGLUE (Lu et al., 2021)	BLEU, CodeBLEU, accuracy

rules. However, the complexity of reasoning depth needs further improvement.

6.2.2 Reasoning Modes

This category explores existing benchmarks used in studies targeting symbolic integrated LLM. These benchmarks are essential for evaluating LLMs' performance across different reasoning models, such as inductive, deductive, and abductive reasoning. Deductive reasoning in symbolic AI-integrated LLMs refers to LLM's ability to apply general rules to specific instances to deduce the conclusion. This includes bAbi-deductive (Weston et al., 2015), Winologic (He et al., 2021), WaNLI (Liu et al., 2022), Bigbench-deduction (Srivastava et al., 2023), Rulebert-Union (Saeed et al., 2021), prOntoQA (Saparov & He, 2022), ReClor (Yu et al., 2020), LogiQA (Liu et al., 2023). Inductive reasoning refers to deducing general conclusions from specific observations. Benchmarks in this category evaluate LLMs' ability to identify patterns to deduce generalised findings. This includes CLUTTR-Systematic (Sinha et al., 2019), Bigbench-logical-args, bAbi-inductive. Abductive reasoning involves forming the most likely explanation for an observation. While less common in benchmarks, some datasets incorporate elements of abductive reasoning. These include LogiGLUE (Luo et al., 2023) and AbductionRule (Young et al., 2021). LOGIGLUE is a benchmark for testing LLM-integrated symbolic reasoning systems (Luo et al., 2023). It contains 24 datasets for deductive, abductive, and inductive reasoning. LOGIGLUE instructs a language model named LogiT5 trained for single-task, multi-task, and chain-of-thought techniques to assess logical reasoning with limited external knowledge (Luo et al., 2023).

6.2.3 Domain-Specific Reasoning

Rapid improvements in approaches to integrating Symbolic AI with LLMs have enhanced domain-specific reasoning, resulting in new pathways for designing reasoning benchmark datasets specific to each domain. This section explores benchmark datasets that existing studies use to assess reasoning abilities in mathematical problems, coding, and algorithms.

Datasets for Mathematical and Physics Reasoning: Mathematical reasoning datasets assess LLMs' capability to handle numerical and logical tasks. Benchmarks include Grade School Math (GSM) Symbolic (Mirzadeh, 2025), MAWPS. These datasets assess LLMs' problem-solving, arithmetic, and logical strengths, requiring sequential reasoning. ARB is another benchmark proposed to address advanced reasoning problems in multiple fields, including math and physics (Sawada et al., 2023). This benchmark provides quantitative analysis covering a high number of short answers and open responses compared to a low number of multiple-choice questions. Another linguistic physical reasoning benchmark in (Zhang et al. 2023) introduces PiLoT (Physics in a Language of Thought), which enhances LLMs with symbolic reasoning by translating natural language into probabilistic programs that generate and simulate physical scenes. This framework enables structured logical inference about physical phenomena beyond simple pattern matching.

Datasets for Coding and Computational Reasoning: Programming and coding-related tasks use symbolic approaches to enhance LLM's logical structure (Deng et al., 2023). Notable benchmarks used in existing studies include Abstraction and Reasoning Corpus (ARC) (Alford, 2021) and CodeXGLUE (Lu et al., 2021). CodeXGLUE evaluates LLMs' ability to assess code intelligence across various programming-related tasks. ARC is a benchmark that can perform few-shot visual reasoning to determine an LLM agent's ability to solve complex tasks. These benchmarks for symbolic-integrated LLMs across various domains highlight the potential to assess improved reasoning in mathematical, algorithmic, and coding tasks.

6.3 Quantitative Benchmark Analysis Across Taxonomy Paradigms

This section provides a quantitative comparison of representative methods. A small subset of benchmarks was selected using strict inclusion criteria. Specifically, a benchmark was included only if it was used by more than one reviewed method, reported with comparable metrics, evaluated under a similar task setting, included a clear baseline or comparator, and could be linked to the proposed taxonomy paradigms. Table 10 summarises the resulting performance gains and highlights preliminary empirical patterns across symbolic-integrated LLM paradigms.

Limitations of Evaluation Metrics Standard metrics including Accuracy, F1, and BLEU are commonly used across benchmarks, but they are limited for evaluating symbolic reasoning in LLMs. Accuracy captures only final answer correctness and cannot distinguish genuine logical reasoning from surface-level pattern matching. BLEU relies on lexical overlap, making it poorly suited to symbolic outputs, where logically equivalent expressions may differ in form. F1, similarly, measures token- or entity-level overlap but does not assess the validity of intermediate reasoning steps in multi-hop inference. These limitations are even more pronounced in neuro-symbolic systems, where current metrics

Table 10 Representative quantitative gains on shared benchmarks, cross-mapped to the proposed taxonomy. Gains are reported only for directly comparable settings within the source papers

Benchmark	Method	Comparator	Reported gain	Taxonomy mapping
ProofWriter	Logic-LM	GPT-4 Chain-of-Thought (CoT)	+11.55 Acc.	Inf; Mod; App; L→S
	LINC	GPT-4 Chain-of-Thought (CoT)	+26 Acc.	Inf; Mod; Alg; L→S
CommonsenseQA	JointLK	QA-GNN	+0.5 official test and +1.02 in-house	FT; Tight; Alg; Hyb
	GreaseLM	prior best LM+KG model	+0.9 Acc.	FT; Tight; Alg; Hyb
OpenBookQA	JointLK	QA-GNN	+2.15 Acc.	FT; Tight; Alg; Hyb
	GreaseLM	QA-GNN	+2.0 Acc.	FT; Tight; Alg; Hyb
ComplexWebQuestions (CWQ)	RoG	UniKGQA	+22.3 Hits@1 and +14.4 F1	FT+Inf; Mod–Tight; App; S→L
	ChatKBQA	previous best	+4 F1, +16 Hits@1, and +4 Acc.	FT; Mod; App; L→S
FOLIO	Logic-LM	GPT-4 Tree-of-Thought (ToT)	+8.1 Acc.	Inf; Mod; App; L→S
	LINC	GPT-4 Tree-of-Thought (ToT)	+3.1 Acc.	Inf; Mod; Alg; L→S

Note: *Inf* inference, *FT* fine-tuning, *Mod* moderately coupled, *App* application-level, *Alg* algorithm-level, *L→S* LLM-to-symbolic, *S→L* symbolic-to-LLM, *Hyb* hybrid

cannot separate the contribution of symbolic components from that of the LLM itself.

Data Contamination Reasoning datasets used for training and instruction tuning may overlap with test examples used for evaluation, leading to inflated performance metrics that do not accurately reflect real-world reasoning capabilities.

The Limited Scope of Logic Types Real-world reasoning often involves a mix of deductive, inductive, and abductive reasoning, yet many benchmarks fail to cover these modes adequately. Additionally, these benchmarks tend to focus only on either PL or FOL, which limits the versatility of LLMs across diverse domains.

Scalability and Complexity in Benchmarks The complexity of extensive and interconnected KGs challenges LLMs, making the tasks more computationally demanding.

Incompleteness and Inconsistency Benchmarks face issues of ambiguity and variability when translating natural language prompts into formal logic. Incomplete data in KGs can hinder reasoning, making it difficult for LLMs to fill knowledge gaps and retrieve facts reliably without relying on probabilistic assumptions.

Relevance and Noise Filtering Another challenge involves filtering out irrelevant information while managing complex entity relations, which can introduce noise in multi-hop reasoning.

Evaluation of Interpretability Development of diverse benchmarks is vital. Current benchmarks often focus on individual, specific reasoning types, limiting the scope of testing. They may not capture the full spectrum of symbolic reasoning across varying complexity levels and modes. Diverse datasets that encompass a broader range of reasoning modes and support complex, multi-step, multi-logic reasoning tasks should be developed. Benchmarks capable of measuring context maintenance across multi-hop KG relationships could improve integration, enhance KG alignment with LLMs, and optimise query efficiency during benchmark planning. Domain-specific benchmarks are essential for designing and improving interpretability.

Hybrid Reasoning and Multimodality For better real-world applicability, more comprehensive benchmarks that combine symbolic logic, KGS, and multimodal data are crucial.

6.4 Recommendations for Future Benchmark Design

The challenges discussed above, particularly the limitations of current evaluation metrics, motivate several recommendations for improving future benchmark design.

1. **Process-level evaluation** Benchmarks should assess the logical validity of intermediate reasoning steps, not only final answer correctness.
2. **Symbolic consistency metrics** Evaluation should test whether outputs are formally entailed by the symbolic component, rather than relying only on string matching.
3. **Ablation-based attribution metrics** Benchmarks should quantify the marginal contribution of the symbolic component, so that gains from symbolic grounding can be distinguished from those arising from the LLM's parametric knowledge.

These recommendations address a central limitation across the reviewed benchmarks. Existing metrics assess output

quality, but offer little insight into the reasoning process or the specific contribution of symbolic integration.

7 Symbolic-LLM Integration Role

Integrating symbolic approaches with LLMs can address neural architectures' limitations while leveraging their strengths. This section discusses symbolic integration's role in enhancing LLM capabilities across various domains. Three critical areas explored in the section include knowledge representation with embeddings, reasoning, planning, and problem-solving.

7.1 Knowledge Representation and Embedding

Symbolic integration can provide structured representations of entities, relations and rules, which may be aligned with continuous embeddings used by an LLM. These embeddings translate logic and the entities and relationships of KGs into continuous vector spaces, capturing the semantic relationships between concepts. Improved semantic relationships can generate contextually relevant knowledge representations. KG-based embedding methods include translational distance and semantic matching models (Ge et al., 2024; Zhang et al., 2020). Advanced embedding techniques use a complex vector space to differentiate symmetric and asymmetric facts (Wang et al., 2021). However, the limited structural connectivity in such approaches often results in incorrect handling of long-tailed relationships and does not enable the prediction of unseen entities (Wang et al., 2019). Considering these challenges, current research is adapting LLMs for enhanced representations of entities and relationships (Wang et al., 2021). A contrastive learning method named LMKE is proposed to address the problem of long-tail entities and improve the learning of embeddings generated by LLMs for KG (Wang et al., 2022). LambdaKG is another LLM-based embedding approach proposed to enhance graph structure representation by concatenating neighbour entity tokens with the triple to feed the sentence into LLMs (Xie et al., 2022). COMET is a framework proposed for dynamically expanding KGs that has demonstrated enhanced performance in reasoning tasks (Bosselut et al., 2020). Another approach utilises GPT-3 to create high-quality knowledge triples to augment existing KGS (Kim et al., 2023), resulting in highly relevant knowledge retrieval. Embedding symbolic knowledge into LLMs can result in high-quality domain-specific responses. KnowBert injects knowledge bases into language models to improve performance on entity-linked tasks (Peters et al., 2019). Symbol-LLM injects symbolic knowledge into LLMs without compromising their general language understanding (Xu et al., 2024).

7.2 Reasoning

Symbolic reasoning refers to manipulating symbolic logic and KGs according to predefined rules and patterns to perform reasoning. LLM-guided symbolic reasoning bridges the gap between the statistical learning capability of LLM and the logic-driven process of the symbolic systems to gain state-of-the-art results on complex reasoning tasks. Symbolic systems excel in breaking down complex problems into step-by-step logical processes. Combining this strength with LLMs enables complex multi-hop reasoning over structured knowledge, such as knowledge graphs or rule-based systems (Yang et al., 2024). Symbolic integration enhances logical consistency by integrating formal logic and symbolic solvers with LLMs. A framework is proposed in Narin (2024) to generate intermediate reasoning steps using LLMs and step verification using symbolic solvers. ProofWriter uses LLM to create proofs verified by automated theorem provers and has achieved state-of-the-art results on several reasoning benchmarks (Tafjord et al., 2021). Symbolically guided approaches are widely used for symbolic verification of LLM outputs to ensure reliability. LLM-generated outputs are verified using symbolic methods, and errors in task planning are reduced (Ding et al., 2023). Such techniques are used to verify the factual accuracy of LLM-generated text using KGs (Liu et al., 2024). KG-based RAG approaches can retrieve entities and relationships pertinent to queries, enabling the RAG system to perform deductive reasoning and coherent explanations (Tan et al., 2024). Existing studies suggest that in LLM-integrated symbolic approaches, LLMs can act as symbolic reasoners capable of performing symbolic tasks in real-world applications (He-Yueya et al., 2023). While symbolic-integrated LLMs possess strong reasoning capabilities, current systems still struggle with high-level multimodal reasoning. It should be noted that LLM reasoning capabilities are not static. Recent advances in structured reasoning strategies such as Chain-of-Thought prompting, tool-augmented inference, and test-time computation scaling have significantly strengthened LLM reasoning performance, partially narrowing the gap that symbolic integration addresses. The motivation for symbolic integration should therefore be understood as complementing and grounding increasingly capable LLM reasoning with verifiable and explainable symbolic knowledge, particularly in high-stakes domains where stronger assurance, verification and traceability are required.

7.3 Planning and Problem-Solving

LLM-integrated symbolic AI approaches can be utilised to solve complex planning problems. LLM-Planner is a system that uses an LLM to generate candidate plans that are refined or validated using symbolic methods (Song et al., 2023). Plansformer uses LLMs to develop highly relevant symbolic plans (Pallagani et al., 2023). Symbolic approaches have effectively solved complex problems by breaking them down into subproblems. Collaborative problem-solving, widely discussed in the literature, is a natural extension in which multiple symbolic–LLM agents tackle decomposed subproblems and coordinate their symbolic reasoning to achieve comprehensive solutions. Logic-LM proposed, and an LLM integrated, a symbolic approach using dedicated symbolic solvers to improve logical problem-solving on complex issues (Pan et al., 2023). Nonetheless, intricate problem-solving strategies face difficulties arising from the token-level constraints inherent in LLM (Luo et al., 2023a). Incorporating symbolic reasoning techniques, such as chain-of-thought (CoT) and tree-of-thought (ToT), can enhance token-level problem-solving by encouraging a clear, step-by-step articulation of thought processes (Luo et al., 2023a).

7.4 Symbolic-Integrated LLM to Address Explainability

Explainability refers to the details and reasons a model provides to make its functioning and decisions clear. Explainability is one dimension of responsible AI and interacts with trustworthiness, robustness, safety, security and the quality of model reasoning. Various approaches are employed to address these challenges to ensure explainability (Hendrycks et al., 2020). Explainability can be examined at the data, model and post-inference levels (Cambria et al., 2024). Data-level explainability involves using features, relevant examples, attention weights, or attributes from the dataset to generate an explanation for why an LLM made a particular prediction. Model-based explainability involves analysing the internal processes of the algorithm, focusing on LLMs' inner layers, embeddings, neurons, and activation functions to provide understanding. Post-inference explainability refers to interpreting the decision-making process after response generation, mainly for evaluating the LLM's performance. Integrating LLMs with symbolic AI has produced advances in reasoning, explainability and evaluation, but they present various challenges, with explainability being a notable concern. Table 11 offers a comparative analysis of different symbolic approaches integrated with LLMs, focusing on multiple aspects of explainability. The factors considered in this study include reasoning, robustness, and ethics. Table 11 provides a structured overview of this analysis. The comparison outlines how logic-based and graph-based integrations with LLMs like GPT and Llama function. It also highlights the limitations of each approach concerning specific explainability aspects and datasets, illuminating challenges researchers face. Common limitations include scalability issues and high dependence on the LLM used. This table also details the evaluation metrics employed to measure these aspects. By analysing the performance and limitations of each approach, this comparison demonstrates how different integration strategies impact the capability and scalability of LLMS. The focus lies on assessing the strengths and weaknesses of each process in terms of explainability. This comparison provides insights into the current state of symbolic integration and helps identify gaps in explainable, symbolic-integrated LLMs. These insights can serve as a foundation for future research development.

8 State-of-the-Art Achievements and Challenges

8.1 State-of-the-Art Achievements

Integrating LLMs with symbolic AI has addressed various challenges, including enhanced reasoning, explainability, and benchmarks.

Methodological Advancements Considering Architectural Paradigms In LLM-to-Symbolic representation, various SOTA approaches discussed in the literature include symbolic-prompt engineering, symbolically guided, instruction-tuned LLMs, schema and ontology-guided LLMs, Program of Thought, incorporating syntactic and semantic parsers and Tree-of-Thoughts. SOTA approaches for LLM-based action schema generation include few-shot prompting (Song et al., 2023), corrective re-prompting (Raman et al., 2022), fine-tuning (Pallagani et al., 2023), agent planning (Li et al., 2024), and hybrid LLM-symbolic methods (Li et al., 2024) integrating predefined logic frameworks, schema libraries (Huang et al., 2025), and automated validation modules. In Symbolic-to-LLM architecture, innovative knowledge injection approaches include encoding symbolic rules into input prompts, embedding logic and KGs into LLMs, ontology-based injection and prompting, constraint-based fine-tuning, Logic-Informed Fine-Tuning with Symbolic Rules, Schema and Template-Based KG Injection, and hybrid approaches for validation or refinement. Predefined templates for symbolic structure conversion to natural language

Table 11 Comparative analysis of approaches targeting explainability

Study	Explainability Factors	Approach	LLM	Symbolic	Dataset	Evaluation	Limitation
(Quan et al., 2024)	Ethics - Correctness and completeness ensured by ethical reasoning	Abductive-deductive framework combining LLMs with symbolic backward-chaining	GPT-3.5-turbo	Logic and rule based formal language through auto formalization (Prolog based)	Ethical NLI benchmarks with commonsense reasoning	Macro-average F1 score at 3rd iteration for hard ethical statements is 58.6% compared to 54.1% for CoT	Unable to provide satisfactory reasoning on complex ethical dilemmas in different cultures because of poor interpretation
(Yang et al., 2023)	Robustness and reasoning	LLM integrated with logic programming named Answer set programming. Where LLM acts as semantic parser to give input for answer set programs.	GPT-3,	Logic programming, Answer set Programming	Several NLP benchmarks, including bAbI, StepGame, CLUTRR, and gSCAN.	GPT-3(d3)+ASP on bAbI gives 99.99% performance	Approach lacks handling errors in the dataset, difficult to write knowledge modules by hand.
(Luo et al., 2023b)	Faithful reasoning	Integrating LLMs with KGs to ensure interpretable reasoning by using structural information captured from KGs by considering relations paths	GPT, Alpaca, LLaMA2, and Flan-T5	KGs	WebQuestionSP (WebQSP) and Complex WebQuestions (CWQ)	For WebQSP, Recall values after integrating with Rog planning for GPT, Alpaca, LLaMA2, and Flan-T5 are 71.60, 74.20, 56.16 and 44.93	Highly dependent on planning, random plans give even worse performance
(Romero et al., 2023)	Robustness	Tightly coupled LLM with cognitive architecture and symbolic rules to enhance the knowledge extraction and integration process	GPT	Symbolic Rules	Prompts		Highly dependent on LLM layer
(Olausson et al., 2023)	Logical reasoning	LLM acts as a semantic parser, to perform deductive inference integrated with FOL expressions	StarCoder+, GPT-3.5, GPT-4	First Order Logic to perform Logical reasoning, and deductive inferring	FOLIO and a balanced subset of ProofWriter	26% higher accuracy with GPT-4 than CoT on ProofWriter (FOLIO)	Only one aspect of logical reasoning is focused, Scalability, computational costs, better reasoning and logic techniques needed

and domain-specific vocabulary-based context-aware translation are the latest breakthroughs for translation.

Innovative Approaches for Overcoming LLM Limitations Integrating LLMs with symbolic AI has resulted in more interpretable and reasoning-oriented systems (Zhu & Sun, 2024). Symbolic AI influences the learning processes of LLMs, leading to enhanced decision-making. Large Language Model - Automated Reasoning Critic (LLM-ARC) is a symbolic-integrated LLM framework that enhances the reasoning capabilities of LLMs by using Answer Set Programming (ASP) (Kalyanpur et al., 2024). It has achieved state-of-the-art accuracy of 88.32% on the FOLIO benchmark (Zhu & Sun, 2024; Ferrag et al., 2025). A central potential benefit of symbolic integration is improved explainability through explicit knowledge, constraints or reasoning traces. Symbol-LLM (Xu et al., 2024) and SymbolicAI (Dinu et al., 2024) are two approaches that use symbolic logic and rules to enhance explainability. Symbolic approaches are well known for learning from limited and unseen data, which can improve learning efficiency in LLMs (Li et al., 2024; Calanzone et al., 2024). This results in better knowledge transfer across domains without retraining the extensive models. Symbolic integration has resulted in enhanced problem-solving, as demonstrated by Logic-LM, which integrates LLMs with symbolic solvers to enhance logical problem-solving capabilities. This integration has improved LLMs' performance from various perspectives, but has increased parameter sizes and long processing times.

Methodological Advancements Considering Integration Strategies The integration of symbolic knowledge

at various stages of LLMs has achieved state-of-the-art results. In the pretraining stage, integration is accomplished through instruction tuning, incorporating symbolic knowledge into model input, or embedding symbolic knowledge within training objectives, all considered breakthroughs. In the inference stage, cutting-edge approaches include symbolic prompting, probing, and retrieval-augmented knowledge fusion. During training and post-training integration, remarkable progress has been made using LLMs as symbolic encoders and decoders for constructing symbolic resources, including knowledge graphs and logic embeddings. In the post-training phase, leading innovations use symbolic structures to update the vast data already learned by LLMs without retraining the model. However, such approaches struggle with variable output quality and additional computational overhead. Strategies for effective knowledge injection remain an open research domain for discussion.

Knowledge Extraction and Validation Approaches Symbolic approaches overcome LLM's challenges of incorporating external knowledge. Prominent recent approaches include retrieval-augmented generation (RAG) and parameter-efficient fine-tuning (PEFT) by encoding KG from node and relation perspectives using KG adapters (Tian et al., 2024). Other advances include hybrid knowledge-validation methods and approaches designed to improve scalability, accuracy and interpretability (De Santis et al., 2024; Zhu et al., 2023). The GSM8K benchmark is widely used to assess mathematical reasoning performance (Cobbe et al., 2021). Based on GSM8K, GSM-Symbolic is designed to determine the variability in LLM performance across different formulations of the same problem, revealing limitations in logical reasoning capabilities (Mirzadeh, 2025). SGP-Bench is another benchmark to evaluate LLMs' semantic understanding and consistency in symbolic graphics programs (Qiu et al., 2024). Further, Graph Neural Networks (GNNs) have continuously evolved and emerged as a robust framework for enhanced learning from graphs (Tang et al., 2024). Cutting-edge approaches such as Graph Neural Networks (GNN)-enhanced LLMs and Graph based Retrieval-Augmented Generation (RAGs) have emerged as pivotal innovations. GNN-enhanced LLMs leverage structured knowledge representations to refine contextual embeddings, allowing models to reason over relational data more effectively. Meanwhile, graph based RAG integrates graph-based retrieval mechanisms to ground generation in retrieved graph evidence, which can reduce unsupported claims when retrieval and graph quality are adequate. Other notable breakthroughs include approaches like Joint Pre-training of Knowledge Graph and Language Understanding, which simultaneously learns from textual and structured knowledge to improve adaptability across domains. These methodologies have demonstrated superior performance on benchmarks such as OpenBookQA and WebQuestionsSP, validating their effectiveness.

8.2 Challenges and Future Directions

Established Practices and Design Patterns Current research lacks systematic examination of the design patterns that govern LLM-symbolic integration, which represents a substantial opportunity for theoretical advancement (d'Avila Garcezd'Avila Garcez & Lamb, 2023). These practices require supporting evidence to validate the integration strategy (De Boer et al., 2023). Comparable empirical and theoretical evidence is needed to guide the selection of symbolic components, including first-order and higher-order logic, symbolic rules, and logical or probabilistic language. While existing approaches exhibit promising results (Peng et al., 2024), the absence of comprehensive pattern analysis limits the field's ability to develop coherent architectural principles. Future research should prioritize the identification and formalization of these patterns to enable more methodical and transferable approaches to neurosymbolic system design.

Benchmarks Another notable challenge is the need for standard benchmarks on symbolic knowledge and to ensure the quality and diversity of datasets through rigorous validation (Acharya et al., 2024). More diverse datasets suitable for different symbolic approaches are also regarded as necessary for evaluation (Jin et al., 2024). Conflict resolution algorithms: A crucial challenge lies in designing conflict resolution algorithms that consistently resolve conflicting information between symbolic modules and LLM-generated responses to ensure consistency (Peng et al., 2024).

Hallucination Detection Symbolic-integrated LLM approaches have recently been employed to reduce AI hallucinations. Most literature on symbolic-integrated LLM approaches targets symbolic components as external knowledge bases to validate LLMs (Ji et al., 2023). Some studies have proposed a generalised and seamless fact-checking model to detect hallucinations (Feng et al., 2023). A useful direction is to classify errors by their origin in the parametric model, symbolic resource or integration interface. Based on the studies reviewed, this work proposes an initial taxonomy of hallucination origins in symbolic-integrated LLM systems, classifying them across three categories. (i) **Parametric hallucinations** originate from the LLM's own weights. The model generates factually incorrect content based on patterns learned during pre-training, independent of any symbolic component (Ji et al., 2023). These are the

most widely discussed in the literature and occur regardless of whether symbolic integration is present. (ii) **Symbolic Hallucinations** originate from incomplete or inconsistent symbolic components such as outdated KGs and conflicting rules (Feng et al., 2023). (iii) **Integration hallucinations** occur at the neural-symbolic interface where an error arises from mapping, retrieval, translation, orchestration or conflict resolution at the neural–symbolic interface, even when the individual components appear locally plausible (Kambhampati et al., 2024; Pan et al., 2023). This three-way classification has direct implications for system design: parametric hallucinations call for symbolic grounding at pre-training or fine-tuning stage, symbolic hallucinations call for KG validation and consistency checking, and integration hallucinations call for interface-level verification mechanisms such as those proposed in LLM-Modulo (Kambhampati et al., 2024) and Logic-LM (Pan et al., 2023). In addition, uncertainty-estimation methods for models, symbolic sources and integration interfaces remain an open research direction across all three categories (Pan et al., 2023).

LLM Knowledge Alteration and Fusion Another challenge involves altering the knowledge within LLMs without retraining, utilising symbolic approaches. Some efforts are underway to modify LLM knowledge (Yao et al., 2023), but the literature also suggests this can cause cascading reactions affecting other relevant understanding (Cohen et al., 2024). Additionally, some LLMs only provide API access; hence, accessing their internal structure and parameters is further complicated. This complicates knowledge fusion during pre-training with symbolic data within the LLM pipeline (Sun et al., 2022). Developing systematic frameworks for knowledge updating and augmentation in both open-source and closed-source LLMs is necessary.

Prompt-Based Reasoning Current state-of-the-art prompting approaches are effective for eliciting task-relevant responses from LLMs (Vatsal & Dubey, 2024). However, these methods may often require revision when assessing LLMs' relational reasoning capabilities using symbolic approaches (Li et al., 2024).

RAG and Evaluation Metrics The literature recognises RAG as a state-of-the-art achievement for refining LLM outputs without retraining (Fan et al., 2024). Concerning symbolic AI, KG-based RAG is acknowledged as another significant success (Peng et al., 2024); however, few approaches target symbolic-logic integration within the RAG pipeline. This gap highlights the need for further research to develop methods that specifically incorporate logic into RAG frameworks. Furthermore, evaluation metrics specific to graph RAG and logic RAG are an important area for future development, especially metrics emphasising reasoning and explainability.

Other Challenges Developing mechanisms for incorporating real-time updates from dynamic KGs during training and fine-tuning (Peng et al., 2024) are few other challenges. Improving the integration of structured reasoning paths without compromising performance remains an area for advancement. LLMs face difficulties with graph linearisation and model optimisation, essential steps for input incorporation into LLMs (Jin et al., 2024). Reducing model complexity and computational resource requirements, and improving the efficiency of LLM–KG integration during training, is an ongoing research focus (Li et al., 2024). The literature contains few investigations of specific adaptations of logic-based rules within the training phases of transformer architectures. Knowledge-infused attention mechanisms and transformer architectures require further research. While symbolic integration enhances LLMs by producing more accurate and interpretable responses, addressing these challenges is vital to unlocking the full potential.

8.3 Conclusion

This study examined the challenges associated with LLMs in high-risk sectors, where explainability is essential for transparency and interpretability. Among the approaches proposed to address these challenges, Neurosymbolic AI offers a promising direction by combining neural learning with symbolic reasoning to support more interpretable and logically grounded outputs. In this context, symbolic integration should be understood not as a replacement for the reasoning capabilities of LLMs, but as a means of complementing them with verifiable and explainable symbolic grounding. Based on the reviewed literature, this survey identified symbolic integration as a promising direction for improving explainability in LLMs and examined a range of approaches that apply symbolic techniques for this purpose. The findings also show a growing research interest in this area. However, tightly coupled or compiled forms of integration between symbolic methods and LLMs remain comparatively underexplored, and practical implementations of symbolic-integrated LLMs are still relatively limited, and many studies remain conceptual, prototype-based or limited to benchmark-scale evaluation.

Investigating architectural models that connect LLM-to-symbolic and symbolic-to-LLM pipelines reveals their potential in improving reasoning, explainability, and integration in AI systems. LLM-to-symbolic pipeline suffers from several challenges, including computational complexity, potential inconsistencies between LLM outputs and symbolic representations, dependency on large, annotated

datasets, domain-specific adoption, sensitivity to prompt engineering, and limitations in scalability to complex symbolic structures. Researchers are exploring various solutions to address these issues, including parallel processing techniques, developing intermediate representation layers to bridge the gap between LLM outputs and symbolic inputs, and utilising adaptive weighting mechanisms in hybrid models. However, further research is needed to create more robust and efficient LLM-to-symbolic pipelines. A symbolic-to-natural language translation pipeline requires additional research in designing robust translation algorithms that handle various symbolic formats and provide context-aware translation. Furthermore, a significant gap that has yet to be explored comprehensively is the design of an efficient integration system to incorporate translated content into LLM prompts or processing pipelines, thereby enhancing LLM responses. In addition to the challenges mentioned earlier, future initiatives involving symbolic-to-LLM pipelines should concentrate on several key areas. These include improving how symbolic context is selected, represented and integrated and creating methodologies for verifying and validating symbolic knowledge. Focusing on these aspects can lead to significant improvements in symbolic-to-LLM pipelines. To summarise, while pipeline models provide some level of simplicity, they can struggle with complex, iterative reasoning, highlighting the need for hybrid models.

Considering the reasoning capabilities of symbolic AI, a recommended direction explored in the literature is to combine the strengths of symbolic AI and LLMs for enhanced explainability. The paper discusses state-of-the-art achievements and challenges in integrating symbolic methods, such as KGs and logic-based reasoning, with LLMs at different lifecycle stages. This article introduces stages for symbolic AI integration with LLMs, offering a structured approach to combining their strengths. The level of interaction between symbolic and LLM components is explored in terms of coupling. The algorithm-level and application-level perspective is analysed. A comparative analysis of existing approaches targeting explainability and symbolic-integrated LLMs is provided. This analysis identifies the strengths, limitations and explanatory mechanisms of each approach to help understand which models offer better explainability, where improvements are needed, and how these systems perform specific tasks. Such an analysis provides valuable insights for future research and development in creating more transparent and reliable AI models. This review provides the research community with a clear roadmap for analysing challenges and achievements in symbolically integrated LLMs, fostering the development of new integration methods and promoting more systematic experimentation and innovation.

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Maneeha Rani is a PhD researcher at the University of Hull, United Kingdom. Her research focuses on neurosymbolic artificial intelligence, large language models, knowledge graphs, retrieval-augmented generation, reasoning, and explainable AI. Her work investigates how symbolic knowledge and neural language models can be combined to improve the transparency, reliability, and interpretability of intelligent systems. She is particularly interested in developing trustworthy AI methods that support structured reasoning, knowledge-grounded question answering, and explainable decision-making across real-world domains.

Bhupesh Kumar Mishra received his PhD in Computing from the University of the West of Scotland, United Kingdom, in 2019. He is currently a Lecturer with the Centre of Excellence for Data Science, Artificial Intelligence, and Modeling at the University of Hull, United Kingdom. His research interests include applied artificial intelligence, optimization techniques, machine learning, large language models, responsible and trustworthy AI, and efficient machine learning pipelines. His work focuses on developing practical, scalable, and ethically grounded AI methods that can be adapted, evaluated, and deployed in real-world contexts.

Dhavalkumar Thakker is Professor of Artificial Intelligence and Internet of Things at the University of Hull, where he serves as the Research Lead for Computer Science and leads the interdisciplinary Centre for Responsible Artificial Intelligence. Professor Thakker's research focuses on Responsible AI, encompassing AI safety, explainability, and fairness, including at the edge, with applications across healthcare, social care, manufacturing, circular economy, and critical infrastructure. He has secured and led over £2M as Principal Investigator and contributed to programmes exceeding £15M, funded by EPSRC, Innovate UK, and the EU. He serves as Associate Editor for several international journals and as an expert reviewer for UKRI, including EPSRC, NERC, and NIHR, particularly on AI and data-driven innovation portfolios. He frequently contributes to major international AI and Semantic Web conferences as a reviewer and organiser.